ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician (MAR TECH) (00379)

USQ Senior Engineering Watchkeeper VIC

NQUAL – XXXX

MITE Course ID's
Residential Phase – 122795
OJPR – 122796

RECORD OF AMENDMENTS

The Training Authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC (A)) is the designated development centre responsible for the maintenance of training. Amendments to Performance Objectives (PO) are to be requested through NTDC (A) and approved by CNPTG. Amendments to Lesson Guidance (LG) are to be approved by the Commanding Officer (CO) of NTDC (A).

Revision	Approval Date	Authority
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FOREWORD

1. This training is based on the specification found at DPGR WEBSITE http://cmp-cpm.mil.ca/en/support/military-personnel/dpgr-index.page.

2. A trainee will achieve the qualification NQUAL when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve the Residential-122795, OJPR-122795 when standards listed in the associated Performance Objectives have been met:

4. The designated TEs for this training include all Naval Fleet Schools, under NPTG.

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Marine Technician Victoria Class Submarine Senior Engineering Watchkeeper (VCS SEWK), XXXX NQUAL. NTDC(A) has designed this QSP for the purpose of enabling courseware development, conducting and evaluating training by Naval Fleet Schools, and conducting validation of training by NPTG.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration. The recommended schedule is based on maximum load. Scheduling cells are to increase/decrease course length based on number of trainees enrolled on the course.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	53
Enabling Checks (ECs)	23
Performance Checks (PCs)	21
Actual Training Time (ATT) Sub-total	**12.1 trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Annex B Table 3.1	
Base Administrative Time (BAT)	2
Training Administrative Time (TAT)	2
Training Travel Time (TTT)	N/A
Physical Fitness (PT)	6
Sub-total	**1.25 trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	0.0 **14 trg days**

REQUIRED COURSE STAFFING

3. Identify key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
Senior Instructor	PO1	VCS SEWK	1	All
Junior Instructor	PO2	VCS SEWK	1	All

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

   a. Maximum – 6;

   b. Minimum – 2; and

   c. Optimum – 4.

PREREQUISITES

5. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

   a. Qualified VCS EWK (NQUAL – TBD) with 500 hours;

   b. RQ PO2 MAR TECH MITE 121982;

   c. USQ PO2 MAR TECH – MECH MITE 122003; and

   d. Recommended by Submarine MAR TECH MOSID 00379 Advisors.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

6. PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

7. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES & LESSON GUIDANCE

PO 001 – Advise OOW on Submarine Marine Engineering Systems as a SEWK

1. **Performance**: Advise OOW on Submarine Marine Engineering Systems as a SEWK.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Assistance;
      (3) Supervision; and
      (4) Victoria Class submarine;

   b. Denied: nil;

   c. Environment: any conditions, day or night, at sea or alongside.

3. **Standard**: The VCS Senior Engineering Watchkeeper shall advise the OOW on Marine Systems Engineering operational capabilities and limitations (CT0010 – CT0030), within and outside of normal design intent to include:

   a. submarine power configuration (PT0010);

   b. submarine propulsion capabilities and limitations (CT0400); and

   c. submarine mechanical systems.

4. **Training Limitations**: nil.

EO 001.01 – Advise OOW on Submarine Power Configuration
01.01.01 – Power Configuration

Conditions		Sequence
Classroom		Start of PO 001
Time	**Classification**	**Instructor**
2 X 50 Minute Periods	Unclassified	VCS SEWK Qualified

Learning Objectives	Method
Advise OOW on Submarines Power Configuration.	Explain – Demo

Scenario	The submarine is at sea and has experienced a partial AC failure and as a result, the VCS SEWK is required to advise the OOW on the submarine's power configuration.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic the SCC, MCC, and HQ1 (ERT) with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor will explain to the students, the importance of the duties and responsibilities of the VCS SEWK during a partial AC failure and why it is so vital to provide recommendations to the OOW in a timely manner. The instructor will hand out PC checklist to students so they can see what is required during scenarios. <u>Explain – Demo</u> (10 min) The instructor will act as the OOW and utilizing one of the scenarios demonstrate what the OOW is expecting to be advised on, and why. <u>Scenario</u> (10 min per student) The instructor will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, providing sound recommendations to the OOW (Instructor) on the submarines power configuration. At the same time, the rest of the class will be fulfilling supporting roles of a VCS watch at sea, including HQ 1 and ERT.

Each student will take a turn advising the OOW on the submarines power configuration, specifically addressing the following items:

- Equipment State and Operational Impacts;
- Motor Generator loading requirements for single Motor Generator operations;
- Considerations for loss of an Auxiliary Power Breaker;
- Abnormal system(s) configurations;
- Quiet states and effects on machinery;
- Quiet Running Bill; and
- Defect Rectification Time.

Quiz (10 min)

After all the scenarios are complete, the instructor will quiz the students on OOW priorities. The instructor will assign one of the power configuration items above to a student, who will have to explain what they would advise the OOW on.

Review and Conclusion (10 min)

The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic the SCC, MCC, and HQ1 (ERT). 3) All reference material.	A1 – A3, A20, A22, B3 – B8, B24, B25, B29, B32

Notes to Instructor

Students should be familiar with the learning objectives by end of session.

EO 001.02 – Advise OOW on Submarine Propulsion Capabilities and Limitations.
001.02.01 – Submarine Propulsion Capabilities and Limitations

Conditions		Sequence
Classroom		After LG 001.01.01
Time	**Classification**	**Instructor**
2 X 50 Minute Periods	Unclassified	VCS SEWK Qualified

Learning Objectives	Method
Advise OOW on Submarine Propulsion Capabilities and Limitations.	Explain – Demo

Scenario	The submarine is at sea and has experienced a propulsion failure and as a result, the VCS SEWK is required to advise the OOW on the submarines propulsion capabilities and limitations.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic the SCC, MCC, and HQ1 (ERT) with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor will explain to the students, the importance of the duties and responsibilities of the VCS SEWK during a propulsion failure and why it is so vital to provide recommendations to the OOW in a timely manner. <u>Demo-Performance</u> (10 min) The instructor will act as the OOW and utilizing one of the scenarios demonstrate what the OOW is expecting to be advised on, and why. <u>Scenario</u> (10 min per student) The instructor will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, providing sound recommendations to the OOW on the submarines propulsion capabilities and limitations. At the same time, the rest of the class will be fulfilling supporting roles of a VCS watch at sea, including HQ 1and ERT.

Each student will take a turn advising the OOW on the submarines propulsion capabilities and limitations, specifically addressing the following items:

- Equipment State and Operational impacts;
- Modes of Emergency Operation for the Main Motor;
- Single Armature;
- Armature Isolation;
- Rotary Isolation Links;
- Contactor Replacement Links; and
- Defect Rectification Time.

Quiz (10 min)

After all the scenarios are complete, the instructor will quiz the students on OOW priorities. The instructor will assign one of the propulsion capabilities and limitations above to a student, who will then explain what they would advise the OOW on, and why.

Review and Conclusion (10 min)

The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic the SCC, MCC, and HQ1 (ERT). 3) All reference material.	A1 – A3, A20, A22, B3 – B8, B24, B25, B29, B32

Notes to Instructor

Students should be familiar with the learning objectives by end of session.

EO 001.03 – Advise OOW on Submarine Mechanical Systems
001.03.01 – Mechanical Systems

Conditions		Sequence
Classroom		After LG 001.02.01
Time	**Classification**	**Instructor**
2 X 50 Minute Periods	Unclassified	VCS SEWK Qualified

Learning Objectives	Method
Advise OOW on Submarines Mechanical Systems.	Explain – Demo

Scenario	The submarine is at sea and has experienced a mechanical systems failure and as a result, the VCS SEWK is required to advise the OOW on the submarines mechanical systems.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic the SCC, MCC, and HQ1 (ERT) with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor will explain to the student's the importance of the duties and responsibilities of the VCS SEWK during a mechanical systems failure and why it is so vital to provide recommendations to the OOW in a timely manner. <u>Demo-Performance</u> (10 min) The instructor will act as the OOW and utilizing one of the scenarios demonstrate what the OOW is expecting to be advised on, and why. <u>Scenario</u> (10 min per student) The instructor will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, providing sound recommendations to the OOW on the submarines mechanical systems. At the same time, the rest of the class will be fulfilling supporting roles of a VCS watch at sea, including HQ 1 and ERT.

	Each student will take a turn advising the OOW on the submarines mechanical systems, specifically addressing the following items: • Discuss Equipment State and Operational Impacts; • Abnormal Systems Configurations; • Quiet States and Effects on Machinery; • Quiet Running Bill; and • Defect Rectification Time. <u>Quiz</u> (10 min) After all the scenarios are complete, the instructor will quiz the students on OOW priorities. The instructor will assign one of the mechanical systems above to a student, who will have to explain what they would advise the OOW on, and why. <u>Review and Conclusion</u> (10 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic the SCC, MCC, and HQ1 (ERT). 3) All reference material.	A1 – A3, A20, A22, B1, B2, B7, B9 – B17,B20 – B23, B26-B27, B34-B36

Notes to Instructor
Students should be familiar with the learning objectives by end of session.

EC 001.03- Advise OOW on Submarine Marine Systems Engineering as the SEWK

Scope	Conditions	Sequence
EO 001.1 – EO 001.03	Classroom	After LG 001.03.01
Time	**Classification**	**Instructor**
6 x 50 Minute Periods	Unclassified	VCS SEWK Qualified

Scenario	The submarine is at sea and has experienced a series of mechanical, electrical, and propulsion systems failures. As a result, the VCS SEWK is required to advise the OOW on the submarines immediate platform capabilities and limitations.
Assessment Steps	EC 001.03 will be based on content from EO 001.01 to EO 001.03. 1) Each student will be assessed as the VCS SEWK while the submarine is at sea and experiences a mechanical, electrical, and propulsion systems failures. As such, the classroom will be arranged in a manner that mimics the SCC, MCC, and HQ1 (ERT) with all the required reference and instructional material. At the same time, the rest of the class will be fulfilling supporting roles of a VCS watch at sea, including HQ 1and ERT. 2) The instructor will provide scenarios, supernumerary personnel, and all reference material listed below. 3) The instructor will act as the OOW. 4) Utilizing a scenario, students will be assessed using checklist EC 001.03. 5) Students must complete EC in 40 min. 6) On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Scenarios. 2) Classroom able to mimic the SCC, MCC, and HQ1 (ERT). 3) All reference material.	A1 – A3, A20, A22, B1, B2, B7, B9 – B17,B20 – B23, B26, B27, B34 – B36

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

Item	EC 001.03 Advise OOW on Submarine Marine Engineering Systems as a VCS SEWK	SAT	UNSAT	Remarks
1. Advise OOW on Submarine Power Configuration				
•	Equipment State and Operational Impacts	☐	☐	
•	Motor Generator loading requirements for single Motor Generator operations	☐	☐	
•	Considerations for loss of an Auxiliary Power Breaker	☐	☐	
•	Abnormal system(s) configurations	☐	☐	
•	Quiet states and effects on machinery	☐	☐	
•	Quiet Running Bill	☐	☐	
•	Defect Rectification Time	☐	☐	
2. Advise OOW on Submarine Propulsion Capabilities and Limitations				
•	Equipment State and Operational impacts	☐	☐	
•	Modes of Emergency Operation for the Main Motor	☐	☐	
•	Single Armature	☐	☐	
•	Armature Isolation	☐	☐	
•	Rotary Isolation Links	☐	☐	
•	Contactor Replacement Links	☐	☐	
•	Defect Rectification Time	☐	☐	
3. Advise OOW on Submarines Mechanical Systems				
•	Discuss equipment state and operational impacts	☐	☐	
•	Abnormal systems configurations	☐	☐	
•	Quiet States and effects on machinery	☐	☐	
•	Quiet Running Bill	☐	☐	
•	Defect Rectification Time	☐	☐	

Score: / 19

Trainee must achieve SATs on all factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 001- Advise OOW on Submarine Marine Engineering Systems as the SEWK

Scope	Conditions	Sequence
PO 001	Classroom	After EC 001.03
Time	**Classification**	**Instructor**
6 x 50 Minute Periods	Unclassified	VCS SEWK Qualified

Scenario	The submarine is at sea and has experienced a series of mechanical, electrical, and propulsion systems failures. As a result, the VCS SEWK is required to advise the OOW on the submarines immediate platform capabilities and limitations.
Assessment Steps	1) PC 001 will be based on content from PO 001. 2) Each student will be assessed as the VCS SEWK while the submarine is at sea and experiences a mechanical, electrical, and propulsion systems failures. As such, the classroom will be arranged in a manner that mimics the SCC, MCC, and HQ1 (ERT) with all the required reference and instructional material. At the same time, the rest of the class will be fulfilling supporting roles of a VCS watch at sea, including HQ 1and ERT. 3) The instructor will provide scenarios, supernumerary personnel, and all reference material listed below. 4) The instructor will act as the OOW. 5) Utilizing a scenario, students will be assessed using the PC 001 checklist. 6) Students must complete PC in 40 min. 7) On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Scenarios. 2) Classroom able to mimic the SCC, MCC, and HQ1 (ERT). 3) All reference material.	A1 – A3, A20, A22, A27, B1 – B17, B20 – B27, B34 – B36

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	PC 001 Advise OOW on Submarine Marine Engineering Systems as a VCS SEWK	SAT	UNSAT	Remarks
1. Advise OOW on Submarine Power Configuration				
•	Equipment State and Operational Impacts	☐	☐	
•	Motor Generator loading requirements for single Motor Generator operations	☐	☐	
•	Considerations for loss of an Auxiliary Power Breaker	☐	☐	
•	Abnormal system(s) configurations	☐	☐	
•	Quiet states and effects on machinery	☐	☐	
•	Quiet Running Bill	☐	☐	
•	Defect Rectification Time	☐	☐	
2. Advise OOW on Submarine Propulsion Capabilities and Limitations				
•	Equipment State and Operational impacts	☐	☐	
•	Modes of Emergency Operation for the Main Motor	☐	☐	
•	Single Armature	☐	☐	
•	Armature Isolation	☐	☐	
•	Rotary Isolation Links	☐	☐	
•	Contactor Replacement Links	☐	☐	
•	Defect Rectification Time	☐	☐	
3. Advise OOW on Submarines Mechanical Systems				
•	Discuss Equipment State and Operational Impacts	☐	☐	
•	Abnormal Systems Configurations	☐	☐	
•	Quiet States and Effects on Machinery	☐	☐	
•	Quiet Running Bill	☐	☐	

• Defect Rectification Time	☐	☐	

Score: /19

Trainee must achieve SATs on all factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 – Advise Incident Controller as a VCS SEWK

1. **Performance:** Advise Incident Controller as a VCS SEWK

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Assistance;
      (3) Supervision; and
      (4) Victoria Class submarine.

   b. Denied: nil.

   c. Environment: any conditions, day or night, at sea or alongside.

3. **Standard**: The VCS Senior Engineering Watchkeeper shall advise the Incident Controller as a member of HQ1 by:

   a. Advising on Immediate Damage Control actions (IT0400);

   b. Advising Power and System Isolations (PT0020);

   c. Providing Recommendations (IT0020); and

   d. Managing VCS SEWK Transition from HQ1 to either HQ2 or HQ3.

4. **Training Limitations**: nil.

EO 002.01 – Advise Incident Controller on Immediate Damage Control Actions
002.01.01 – Immediate Damage Control Actions

Conditions		Sequence
Classroom		After PC 001
Time	**Classification**	**Instructor**
1 X 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturer – XO/EO Post IC

Learning Objectives	Method
Advise Incident Controller on immediate Damage Control Actions during EOP's.	Explain – Demo

Scenario	The submarine is at sea and has experienced an EOP and as a result, the VCS SEWK is to advise the Incident Controller on immediate damage control actions to be taken.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic HQ1, ERT, and the Outstations with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor / Guest Lecturer will explain to the student's the importance of the VCS SEWK advising the Incident Controller on immediate damage control actions during EOP's. The instructor will hand out PC checklist to students so they can see what is required during scenarios. <u>Demo-Performance</u> (5 min) The instructor / Guest Lecturer will act as the Incident Controller and utilizing one of the scenarios, demonstrate what the Incident Controller is expecting to be advised on, and why. <u>Scenario</u> (5 min per student) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, providing sound recommendations to the Incident Controller on immediate damage control actions to be taken during EOP's. At the same time, the rest of the class will be fulfilling supporting roles of members of the ERT and Outstations.

	Each student will take a turn advising the Incident Controller on immediate damage control actions to be taken during EOP's, specifically addressing the following items: • Type of Emergency; • Were immediate actions successful; • Class of fire (if applicable); • Risk of Fire (if applicable); and • Whether or not confinement actions have occurred. <u>Review and Conclusion</u> (5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic HQ1, ERT, and the Outstations. 3) All reference material. 4) Guest Lecturer – XO / EO Post IC.	A1 – A3, A36

Notes to Instructor

Students should be familiar with the learning objectives by end of session.

EO 002.02 – Advise Incident Controller on Power and System Isolations
002.02.01 – Power and System Isolations

Conditions		Sequence
Classroom		After LG 002.01.01
Time	**Classification**	**Instructor**
4 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturer – XO/EO Post IC

Learning Objectives	Method
Advise Incident Controller on Submarine Power and Systems isolations during EOP's.	Explain – Demo

Scenario	The submarine is at sea and has experienced an EOP and as a result, the SEWK is to advise the Incident Controller on power and system isolations.
Activity Steps	Set up (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic HQ1, ERT, and the Outstations with all the required reference and instructional material. Introduction (30 min) The instructor / Guest Lecturer will explain to the student's the importance of the VCS SEWK in HQ1. This includes the VCS SEWK utilizing their experience, critical thinking, along with HQ1 documentation, developing a sound plan to advise the Incident Controller on Submarine Power and Systems isolations during EOP's. Demo-Performance (20 min) The instructor / Guest Lecturer will act as the Incident Controller and utilizing one of the scenarios, demonstrate what the Incident Controller is expecting to be advised on and why. Scenario (20 min) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, developing a plan, and providing sound recommendations on advising the Incident Controller on Submarine Power and Systems isolations during EOP's. At the same time, the rest of the class will be fulfilling supporting roles as members of the ERT and Outstations.

	Each student will take a turn developing a plan and advising the Incident Controller on Submarine Power and Systems isolations during EOP's. Students will be expected to utilize their experience and supporting documentation to develop an isolation plan and restoration plan for presentation to the Incident Controller. <u>Quiz</u> (20 min) After all the scenarios are complete, the instructor will quiz the students on Incident Controller priorities. The instructor will present a challenging scenario to the class, whom together will have to explain what they would advise the Incident Controller on. <u>Review and Conclusion</u> (5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	**References**
1) Scenarios. 2) Classroom able to mimic HQ1, ERT, and the Outstations. 3) All reference material. 4) Guest Lecturer – XO / EO Post IC.	A1 – A3, A36

Notes to Instructor
Students should be familiar with the learning objectives by end of session.

EO 002.03 – Provide Recommendations to the Incident Controller
002.03.01 – Recommendations to the Incident Controller

Conditions		Sequence
Classroom		After LG 002.02.01
Time	**Classification**	**Instructor**
2 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturer – XO/EO Post IC

Learning Objectives	Method
Provide Recommendations to the Incident Controller during EOP's	Explain – Demo

Scenario	The submarine is at sea and has experienced an EOP and as a result, the SEWK is to provide recommendations to the Incident Controller.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic HQ1, ERT, and the Outstations with all the required reference and instructional material. <u>Introduction</u> (10 min) The instructor / Guest Lecturer will explain to the student's the importance of the VCS SEWK providing thorough recommendations (based on the plan from LG 002.02.01) to the Incident Controller during EOP's. <u>Demo-Performance</u> (10 min) The instructor / Guest Lecturer will act as the Incident Controller and based on the plan from LG 002.02.01, utilizing one of the scenarios demonstrate what recommendations the Incident Controller is expecting, and why. <u>Scenario</u> (10 min per student) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT. Based on the plan from LG 002.02.01, the student will provide sound recommendations to the Incident Controller on actions to be taken during EOPs. At the same time, the rest of the class will be fulfilling supporting roles of members of the ERT and Outstations.

	Each student will take a turn, making recommendations to the Incident Controller EOP's from the plan in LG 002.02.01, specifically addressing the following items: • Provide the best plan to Float, Move, Fight IAW with Command priorities; • Provide all compartment hazards; • Provide a list of adjacent compartment rounds; and • Dispatch member(s) of ERT to support the On Scene Coordinator and conduct on scene damage assessment. <u>Quiz</u> (10 min) After all the scenarios are complete, the instructor will quiz the students on making recommendations to the Incident Controller. The instructor will assign one of the items to a student, who will be challenged on their recommendations to the Incident Controller. <u>Review and Conclusion</u> (5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic HQ1, ERT, and the Outstations. 3) All reference material. 4) Guest Lecturer – XO / EO Post IC.	A1 – A3, A36

Notes to Instructor

Students should be familiar with the learning objectives by end of session.

EO 002.04 – Manage VCS SEWK Transition from HQ1 to either HQ2 or HQ3
002.04.01 – Transition from HQ1 to either HQ2 or HQ3

Conditions		Sequence
Classroom		After LG 002.03.01
Time	**Classification**	**Instructor**
2 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturer – XO/EO Post IC

Learning Objectives	Method
Manage VCS SEWK Transition from HQ1 to either HQ2 or HQ3	Explain – Demo Three Stars

Scenario	The submarine is at sea closed up at emergency stations responding to an EOP when HQ1 is forced to evacuate to HQ2.
Activity Steps	Set up (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic HQ1, ERT, and the Outstations with all the required reference and instructional material. Introduction (10 min) The instructor / Guest Lecturer will explain to the student's the importance of the VCS SEWK being able to transition from HQ1 to either HQ2 (Motor Room) or HQ3 (WSC) in both a seamless and timely manner. Demo-Performance (10 min) The instructor / Guest Lecturer will act as the Incident Controller and utilizing one of the scenarios, demonstrate what the procedures are when the order is given to evacuate from HQ1 to either HQ2 or HQ3, and what is expected from the SEWK, and why. Scenario (10 min) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, describing the actions to be taken when ordered to evacuate from HQ1, to either HQ2 or HQ3. At the same time, the rest of the class will be fulfilling supporting roles of members of the ERT and Outstations.

	Each student will take a turn, describing the actions to be taken when ordered to evacuate from HQ1, to either HQ2 or HQ3, specifically addressing the following items: • When to relocate to either HQ2 or HQ3; • Quick and uniform establishing of communications; • Seamless Battle Rhythm; and • When to return to HQ1. <u>Quiz</u> (10 min) After all the scenarios are complete, the instructor will quiz the students on the actions to be taken when ordered to evacuate from HQ1, to either HQ2 or HQ3. The instructor will ask each student what two items they think are most important, and why. <u>Review and Conclusion</u> (5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic HQ1, ERT, and the Outstations. 3) All reference material. 4) Guest Lecturer – XO / EO Post IC.	A1 – A3, A36

Notes to Instructor
Students should be familiar with the learning objectives by end of session.

EC 002.04 – Advise HQ1 Incident Controller as a VCS SEWK

Scope	Conditions	Sequence
EO 002.01 – EO 002.04	Classroom	After LG 002.04.01
Time	**Classification**	**Instructor**
6 x 50 Minute Periods	Unclassified	VCS SEWK Qualified

Scenario	The submarine is at sea and has experienced an EOP and as a result, the VCS SEWK is to advise the Incident Controller on Immediate Damage Control Actions, Power and Systems Isolations, recommendations with a through plan, and managing the transition to either HQ2 or HQ3 when forced to evacuate from HQ1.
Assessment Steps	1) EC 002.04 will be based on content from EO 002.01 to EO 002.04. 2) Each student will be assessed as the VCS SEWK during an EOP in HQ1. As such, the classroom will be arranged in such a manner that the student will be sitting at a desk to mimic HQ 1 with all the applicable references. At the same time, other students not being assessed will act as members of the Outstations and the ERT. 3) The instructor will provide scenarios, supernumerary personnel, and all reference material listed below. 4) The instructor will act as the Incident Controller. 5) Utilizing a scenario, students will be assessed using checklist EC 002.04. 6) Students must complete EC in 40 min. 7) On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Scenarios. 2) Classroom able to mimic HQ1, ERT, and the Outstations. 3) Supernumerary personnel. 4) All reference material.	A1 – A3, A36

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

Item	EC 002.04 Advise Incident Controller on Submarine Marine Engineering Systems as a VCS SEWK	SAT	UNSAT	Remarks
1. Advise Incident Controller on Immediate Damage Control Actions				
•	Type of Emergency	☐	☐	
•	Were immediate actions successful	☐	☐	
•	Class of fire (if applicable)	☐	☐	
•	Risk of Fire (if applicable)	☐	☐	
•	Whether or not confinement has occurred	☐	☐	
2. Advise Incident Controller on Power and System Isolations				
•	What immediate actions have occurred via OSC	☐	☐	
•	Submarine power and system isolations	☐	☐	
•	Develop an isolation plan and present to IC	☐	☐	
3. Provide Recommendations to the Incident Controller				
•	Best plan to Float, Move, Fight IAW with Command priorities	☐	☐	
•	Compartment hazards	☐	☐	
•	List of adjacent compartment rounds	☐	☐	
•	Dispatch member(s) of ERT to support the OSC and conduct on scene damage assessment	☐	☐	
4. Manage VCS SEWK Transition from to HQ2 and HQ3				
•	When to relocate from HQ1 to either HQ2 or HQ3	☐	☐	
•	Establish Communications	☐	☐	
•	Seamless Battle Rhythm	☐	☐	
•	When to return to HQ1	☐	☐	

Score: /16

Trainee must achieve SATs on all factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 002 – Advise Incident Controller as a VCS SEWK

Scope	Conditions	Sequence
PO 002	Classroom	After EC 002.04
Time	**Classification**	**Instructor**
6 x 50 Minute Periods	Unclassified	Senior Instructor

Scenario	The submarine is at sea and has experienced an EOP and as a result, the VCS SEWK is to advise the Incident Controller on Immediate Damage Control Actions, Power and Systems Isolations, recommendations with a through plan, and managing the transition to either HQ2 or HQ3 when forced to evacuate from HQ1.
Assessment Steps	1) PC 002 will be based on content from PO 002. 2) Each student will be assessed as the VCS SEWK during an EOP in HQ1. As such, the classroom will be arranged in such a manner that the student will be sitting at a desk to mimic HQ 1 with all the applicable references. At the same time, other students not being assessed will act as members of the Outstations and the ERT. 3) The instructor will provide scenarios, supernumerary personnel, and all reference material listed below. 4) The instructor will act as the Incident Controller. 5) Utilizing a scenario, students will be assessed using checklist PC 002. 6) Students must complete PC in 40 min. 7) On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Scenarios 2) Classroom able to mimic HQ1, ERT, and the Outstations 3) Supernumerary personnel 4) All reference material	A1-A3, A36

Notes to Instructor
Nil

Limitations	N/A
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	PC 002 Advise Incident Controller on Submarine Marine Engineering Systems as a VCS SEWK	SAT	UNSAT	Remarks
1. Advise Incident Controller on Immediate Damage Control Actions				
•	Type of Emergency	☐	☐	
•	Were immediate actions were successful	☐	☐	
•	Class of fire (if applicable)	☐	☐	
•	Risk of Fire (if applicable)	☐	☐	
•	Whether or not confinement has occurred	☐	☐	
2. Advise Incident Controller on Power and System Isolations				
•	what immediate actions have occurred via OSC	☐	☐	
•	submarine power and system isolations	☐	☐	
•	develop an isolation plan and present to IC	☐	☐	
3. Provide Recommendations to the Incident Controller				
•	Best plan to FLOAT, MOVE, FIGHT IAW with Command priorities	☐	☐	
•	Compartment hazards	☐	☐	
•	List of adjacent compartment rounds	☐	☐	
•	Dispatch member(s) of ERT to support the OSC and conduct on scene damage assessment	☐	☐	
4. Provide Recommendations to the Incident Controller				
•	When to relocate from HQ1 to either HQ2 or HQ3	☐	☐	
•	Establish Communications	☐	☐	
•	Seamless Battle Rhythm	☐	☐	
•	When to return to HQ1	☐	☐	

Score: /16

Trainee must achieve SATs on all factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 – Supervise Victoria Class Submarine Maintenance Activities as VCS SEWK

1. **Performance**: Supervise Victoria Class Submarine Maintenance Activities as a VCS SEWK.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Assistance;
      (3) Supervision; and
      (4) Victoria class submarine;

   b. Denied: nil.

   c. Environmental: any conditions, day or night, at sea or alongside.

3. **Standard**: The VCS Senior Engineering Watchkeeper shall supervise Victoria Class Submarine maintenance activities IAW references A1 – A3, A21 – A25, B1 – B36 by:

   a. Managing Unsatisfactory Condition Reports (UCR) (AFT0860);

   b. Supervising Defence Resource Management Information System (DRMIS) (AFT0400);

   c. Responding to Engineering Change (EC) proposals (AFT0450);

   d. Managing Pre Installation Failure (PIF) reports (AFT0500); and

   e. Analyzing Equipment Health Monitoring Data (AFT0870).

4. **Training Limitations**: nil.

EO 003.01 – Respond to Engineering Change Proposals
003.01.01 – Engineering Change Proposals

Conditions		Sequence
Classroom		After PC 002
Time	**Classification**	**Instructor**
1 X 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturers – FTA / NETE / FMF

Learning Objectives	Method
Respond to Engineering Change (EC) proposals.	Explain – Demo

Scenario	The VCS SEWK has received an EC Proposal and is to verify its adoptability to a Victoria Class Submarine.
Activity Steps	<u>Set up</u> (2.5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that has full DWAN access with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor / Guest Lecturer will explain to the student's the importance of the VCS SEWK being able to Respond to Engineering Change (EC) proposals. <u>Demo-Performance</u> (5 min) The instructor / Guest Lecturer will act as the MSEC and utilizing one of the scenarios, demonstrate what the procedures are when the EC proposal is received from NDHQ – SM, and what is expected from the SEWK, and why. <u>Scenario</u> (5 min per student) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, describe the actions to be taken when an EC proposal is received from NDHQ – SM. Each student will take a turn, describing the actions to be taken when an EC proposal is received from NDHQ – SM.

	Quiz (5 min) After all the scenarios are complete, the instructor will quiz the students on the actions to be taken when EC proposal is received from NDHQ – SM. The instructor will ask each student what items they think are most important, and why. Review and Conclusion (2.5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom. 3) All reference material. 4) DWAN. 5) Guest Lecturer – VCS MSEC. 6) Guest Lecturers – FTA / NETE / FMF.	A1 – A3, A21 – A23-A25, B1 – B36

Notes to Instructor
Students should be familiar with each of the learning objectives by end of lectures.

EO 003.02 – Respond to Equipment Health Monitoring Data
003.02.01 – Equipment Health Monitoring Data

Conditions		Sequence
Classroom		After LG 003.01.01
Time	**Classification**	**Instructor**
1 X 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturers – FTA / NETE / FMF

Learning Objectives	Method
Analyzing Equipment Health Monitoring Data.	Explain – Demo

Scenario	The VCS SEWK has received a Vibration Analysis Reading for a Cooling Pump and is required to interpret the data and make recommendations to the MSEC.
Activity Steps	<u>Set up</u> (2.5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that has full DWAN access with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor / Guest Lecturer will explain to the student's the importance of the VCS SEWK being able to Analyze Equipment Health Monitoring Data and make recommendations to the MSEC. <u>Demo-Performance</u> (5 min) The instructor / Guest Lecturer will act as the MSEC and utilizing one of the scenarios, demonstrate what the procedures are when analyzing equipment health monitoring data, and what recommendations are expected from the SEWK, and why. <u>Scenario</u> (5 min per student) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, describe the actions to be taken when analyzing equipment health monitoring data. Each student will take a turn, describing the actions to be taken when analyzing equipment health monitoring data.

	Quiz (5 min) After all the scenarios are complete, the instructor will quiz the students on the actions to be taken when analyzing equipment health monitoring data. The instructor will ask each student what items they think are most important, and why. Review and Conclusion (2.5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom. 3) All reference material. 4) DWAN. 5) Guest Lecturer – VCS MSEC. 6) Guest Lecturers – FTA / NETE / FMF	A1 – A3, A21 – A25, B1 – B36

Notes to Instructor
Students should be familiar with each of the learning objectives by end of lectures.

EO 003.03 – Supervise Defence Resource Management Information System
003.03.01 – Defence Resource Management Information

Conditions		Sequence
Classroom		After LG 003.02.01
Time	**Classification**	**Instructor**
7 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturers – FTA / NETE / FMF

Learning Objectives	Method
Supervising Defense Resource Management Information System (DRMIS).	Interactive Lecture/ Explain – Demo

Scenario	The MSE Dept. has replaced a First Level System Valve that requires closure of both the ND/N1.
Activity Steps	Set up (10 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that has full DWAN access with all the required reference and instructional material. Introduction (10 min) The instructor / Guest Lecturer will explain to the students the importance of the VCS SEWK being able to Supervise the Defense Resource Management Information System. Demo-Performance (45 min) The instructor / Guest Lecturer will act as the MSEC and utilizing one of the scenarios, demonstrate what the procedures are when Supervising Defense Resource Management Information System, and what is expected from the SEWK, and why. Scenario (45 min per student) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, describing the actions to be taken when Supervising Defense Resource Management Information System. Each student will take a turn, describing the actions to be taken when supervising Defense Resource Management Information System.

	Quiz (10 min) After all the scenarios are complete, the instructor will quiz the students on the actions to be taken when EC proposal is received from NDHQ – SM. The instructor will ask each student what items they think are most important, and why. Review and Conclusion (5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom. 3) All reference material. 4) DWAN. 5) Guest Lecturer – VCS MSEC. 6) Guest Lecturers – FTA / NETE / FMF.	A1 – A3, A21 – A25, A37, A38, B1 – B36

Notes to Instructor

Students should be familiar with each of the learning objectives by end of lectures.

EO 003.04 – Manage Marine Systems Engineering Reports
003.04.01 – Marine Systems Engineering Reports

Conditions		Sequence
Classroom		After LG 003.03.01
Time	**Classification**	**Instructor**
1 X 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturers – FTA / NETE / FMF

Learning Objectives	Method
Managing Unsatisfactory Condition Reports (UCR) and Pre Installation Failure (PIF) reports.	Interactive Lecture

Scenario	The MSE department has received an unserviceable part from the Supply System / Chain.
Activity Steps	<u>Set up</u> (2.5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that has full DWAN access with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor / Guest Lecturer will explain to the student's the importance of the VCS SEWK being able to manage UCR's and PIF reports. <u>Demo-Performance</u> (5 min) The instructor / Guest Lecturer will act as the MSEC and utilizing one of the scenarios, demonstrate what the procedures are when managing UCR's and PIF reports, and what is expected from the SEWK, and why. <u>Scenario</u> (5 min) The instructor / Guest Lecturer will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, describing the actions to be taken when managing UCR's and PIF reports. Each student will take a turn, describing the actions to be taken managing UCR's and PIF reports.

	Quiz (5 min) After all the scenarios are complete, the instructor will quiz the students on the actions to be taken when managing UCR's and PIF reports. The instructor will ask each student what items they think are most important, and why. Review and Conclusion (2.5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom. 3) All reference material. 4) DWAN. 5) Guest Lecturer – VCS MSEC. 6) Guest Lecturers – FTA / NETE / FMF.	A1 – A3, A21 – A25, B1 – B36

Notes to Instructor
Students should be familiar with each of the learning objectives by end of lectures.

EC 003.04 – Supervise Victoria Class Submarine Maintenance Activities as a VCS SEWK

Scope	Conditions	Sequence
EO 003.02 – EO 003.04	Classroom	After LG 003.04.01
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturers – FTA / NETE / FMF

Scenario	The submarine is alongside in a work period and is about to replace a part in a first level system. When the part arrives it is found to be unserviceable and as a result, the SEWK is required to raise a PIF as well as a UCR. Upon receipt of the servable part, an N1/ND needs to be properly closed. This means filling out forms in their entirety IAW Ref A37 and A39.
Assessment Steps	Instructions Given to Trainee: 1. EC 003.04 will be a written test based on content from EO 003.02 – EO 003.04. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. All References will be provided. 9. There will be 10 questions that will involve filling out forms, OQE, First Level Documentation, Dismantling sterilized items in DRMIS and closing out N1/NDs.

	10. Pass mark is 70% 11. You will have 1 x 40 min period to complete this exam. 12. Do you have any questions before you start? 13. On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Classroom. 2) Exam. 3) All reference material. 4) DWAN. 5) DS Solution.	A1 – A3, A8 – A10, A21 – A25, A34, A37, A38, B1 – B36

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

PC 003 – Supervise Victoria Class Submarine Maintenance Activities as a VCS SEWK

Scope	Conditions	Sequence
PO 003	Classroom	After EC 003.04
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturers – FTA / NETE / FMF

Scenario	The submarine is alongside in a work period and is about to replace a part in a first level system. When the part arrives it is found to be unserviceable and as a result, the SEWK is required to raise a PIF as well as a UCR. Upon receipt of the servable part, an N1/ND needs to be properly closed. This means filling out forms in their entirety.
Assessment Steps	Instructions Given to Trainee: 1. PC 003 will be a written test based on content from PO 003. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. All References will be provided. 9. There will be 10 questions that will involve filling out forms, OQE, First Level Documentation, Dismantling sterilized items in DRMIS and closing out N1/NDs.

	10. 10 Pass mark is 70% 11. You will have 1 x 40 min period to complete this exam. 12. Do you have any questions before you start? 13. On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	**References**
1) Classroom. 2) Exam. 3) All reference material. 4) DWAN. 5) DS Solution.	A1 – A3, A8 – A10, A21 – A25, A34, A37, A38, B1 – B36
Notes to Instructor	
Nil	

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PO 004 – Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK

1. **Performance**: Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Assistance;
      (3) Supervision; and
      (4) Victoria Class Submarine.

   b. Denied: nil.

   c. Environment: any conditions, day or night, at sea or alongside.

3. **Standard**: The VCS Senior Engineering Watchkeeper shall supervise Fueling and Defueling (OT0130) on a Victoria Class Submarine IAW references A1 – A7, A22, A30, B7, B27 by:

   a. Supervising Submarine Fueling by:

      (1) Preparing and/or Reviewing Fueling orders;
      (2) Verifying system line up, protocol and resources;
      (3) Conducting safety brief;
      (4) Conducting fuel quality testing (OT0190); and
      (5) Securing from fueling.

   b. Supervising Submarine Defueling by:

      (1) Preparing and/or Reviewing Fueling orders
      (2) Verify system line up, protocol and resources;
      (3) Conduct safety brief; and
      (4) Secure from defueling.

4. **Training Limitations**: nil.

EO 004.01 – Verify Fueling and Defueling System Line Up
004.01.01 – System Line Up

Conditions		Sequence
Classroom		After PC 003
Time	**Classification**	**Instructor**
2 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturers – VCS MSEC, Formation Safety and Environment

Learning Objectives	Method
Verify Victoria Class Submarine Fueling and Defueling System Line Up.	Explain – Demo

Scenario	The submarine is alongside and is preparing to fuel / defuel.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic the VCS Oil Fuel System with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor / Guest Lecturer will explain to the students the importance of the duties and responsibilities of the VCS SEWK during Oil Fuel System, specifically, setup prior to fueling and defueling, as well as the environmental consequences. <u>Demo-Performance</u> (15 min) The instructor / Guest Lecturer will act as the VCS SEWK and utilizing one of the scenarios, demonstrate what is expected of the VCS SEWK prior to embarking fuel, and why. <u>Scenario</u> (10 min per student) The instructor will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, ensuring that: • Preparing and/or Reviewing Fueling orders; and • Verify system line up, protocol and resources IAW SEMS/SOP's.

	Quiz (10 min) After all the scenarios are complete, the instructor / Guest Lecturer will quiz the students on SEMS/SOP's. Review and Conclusion (5 min) The instructor / Guest Lecturer will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic a VCS Oil Fuel System. 3) All reference material. 4) Guest Lecturer – VCS MSEC. 5) Guest Lecturer – Formation Safety and Environment.	A1 – A7, A22, A30, B7, B27

Notes to Instructor
Students should be familiar with each learning objectives by end of lectures.

EO 004.02 – Conduct Fueling and Defueling Safety Brief
004.02.01 – Safety Brief

Conditions		Sequence
Classroom		After LG 004.01.01
Time	**Classification**	**Instructor**
1 X 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturers – VCS MSEC, Formation Safety and Environment

Learning Objectives	Method
Conduct Victoria Class Submarine Fueling and Defueling Safety Brief.	Explain – Demo

Scenario	The submarine is alongside and is preparing to fuel / defuel.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic the VCS Oil Fuel System with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor / Guest Lecturer will explain to the students the importance of the duties and responsibilities of the VCS SEWK during the conduct of the safety brief, prior to fueling and defueling and the environmental consequences. <u>Demo-Performance</u> (5 min) The instructor will act as the VCS SEWK and utilizing one of the scenarios demonstrate what is expected of the VCS SEWK during the safety brief, and why. <u>Scenario</u> (5 min per student) The instructor will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, ensuring that a proper safety brief is conducted to the Fueling Party. <u>Review and Conclusion</u> (5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic a VCS Oil Fuel System. 3) All reference material. 4) Guest Lecturer – VCS MSEC. 5) Guest Lecturer – Formation Safety and Environment.	A1 – A7, A22, A30, B7, B27

Notes to Instructor
Students should be familiar with each learning objectives by end of lectures.

EO 004.03 – Secure from Fueling and Defueling
004.03.01 – Secure Fuel System

Conditions		Sequence
Classroom		After LG 004.02.01
Time	**Classification**	**Instructor**
1 X 50 Minute Period (Classroom and Tour)	Unclassified	VCS SEWK Qualified Guest Lecturers – VCS MSEC, Formation Safety and Environment

Learning Objectives	Method
Secure from Victoria Class Submarine Fueling and Defueling.	Interactive Lecture

Scenario	The submarine is alongside and is preparing to secure from fueling and defueling.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class that is able to mimic the VCS Oil Fuel System with all the required reference and instructional material. <u>Introduction</u> (5 min) The instructor / Guest Lecturer will explain to the students the importance of the duties and responsibilities of the VCS SEWK while securing from fueling and defueling, and the environmental consequences. <u>Demo-Performance</u> (5 min) The instructor will act as the VCS SEWK and utilizing one of the scenarios demonstrate what is expected of the VCS SEWK while securing from fueling and defueling, and why. <u>Scenario</u> (5 min per student) The instructor will have a series of scenarios available for use. A different scenario will be given to each student who will act as the VCS SEWK, honing their skills as an experienced OPCERT, ensuring that the submarine is properly and safely secured from fueling IAW SEMS/SOP's.

	Review and Conclusion (5 min) The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) Classroom able to mimic a VCS Oil Fuel System. 3) All reference material. 4) Guest Lecturer – VCS MSEC. 5) Guest Lecturer – Formation Safety and Environment.	A1 – A7, A22, A30, B7, B27

Notes to Instructor
Students should be familiar with each learning objectives by end of lectures.

EC 004.03 – Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK

Scope	Conditions	Sequence
EO 004.01 – EO 004.03	Classroom	After LG 004.03.01
Time	**Classification**	**Instructor**
6 x 50 Minute Periods	Unclassified	VCS SEWK Qualified

Scenario	The submarine is berthed alongside the fueling jetty and is making preparations to fuel and defuel #1 Oil Fuel Group.
Assessment Steps	1) EC 004 will be based on content from EO 004.01 – EO 004.03. 2) Each student will be assessed as the VCS SEWK supervising fueling and defueling on a Victoria Class submarine. As such, the classroom will be arranged in such a manner that the student will be sitting at a desk to mimic their position as the fueling – defueling supervisor. At the same time, other students not being assessed will act as members of the Fueling Party. 3) The instructor will provide scenarios, supernumerary personnel, and all reference material listed below. 4) The instructor will act as the MSEC. 5) Utilizing a scenario, students will be assessed using checklist EC 004. 6) Students must complete the EC in 40 min. 7) On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Scenarios 2) Classroom able to mimic a VCS Oil Fuel System 3) All reference material 4) Guest Lecturer – VCS MSEC 5) Guest Lecturer – Formation Safety and Environment	A1 – A7, A22, A30, B7, B27

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

Item	EC 004.03 Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK	SAT	UNSAT	Remarks
1. Verify Fueling and Defueling System Line Up				
• Preparing and Reviewing Fueling orders • Verify system line up, protocol and resources IAW SEMS/SOP's		☐	☐	
2. Conduct Fueling and Defueling Safety Brief				
• Conduct Safety Brief		☐	☐	
3. Secure from Fueling and Defueling				
• Securing from fueling IAW SEMS/SOP's		☐	☐	

Score: /3

Trainee must achieve SATs on all factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PC 004 – Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK

Scope	Conditions	Sequence
PO 004	Classroom	EC 004.03
Time	**Classification**	**Instructor**
6 x 50 Minute Periods	Unclassified	VCS SEWK Qualified

Scenario	The submarine is berthed alongside the fueling jetty and is making preparations to fuel and defuel #1 Oil Fuel Group.
Assessment Steps	1) PC 004 will be based on content from PO 004. 2) Each student will be assessed as the VCS SEWK supervising fueling and defueling on a Victoria Class submarine. As such, the classroom will be arranged in such a manner that the student will be sitting at a desk to mimic their position as the fueling – defueling supervisor. At the same time, other students not being assessed will act as members of the Fueling Party. 3) The instructor will provide scenarios, supernumerary personnel, and all reference material listed below. 4) The instructor will act as the MSEC. 5) Utilizing a scenario, students will be assessed using checklist PC 004. 6) Students must complete the PC in 40 min. 7) On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Scenarios. 2) Classroom able to mimic a VCS Oil Fuel System. 3) All reference material. 4) Guest Lecturer – VCS MSEC. 5) Guest Lecturer – Formation Safety and Environment.	A1 – A7, A22, A30, B7, B27

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	PC 004 Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK	SAT	UNSAT	Remarks
1. Verify Fueling and Defueling System Line Up				
• Preparing and/ or Reviewing Fueling orders • Verify system line up, protocol and resources IAW SEMS/SOP's		☐	☐	
2. Conduct Fueling and Defueling Safety Brief				
• Conduct safety brief		☐	☐	
3. Secure from Fueling and Defueling				
• Securing from fueling IAW SEMS/SOP's		☐	☐	

Score: /3

Trainee must achieve SATs on all factors to pass this PC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 – Manage Compliance with the SUBSAFE Policy on a Victoria Class Submarine as a SEWK

1. **Performance**: Manage Compliance with the SUBSAFE Policy on a Victoria Class Submarine as a SEWK

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Assistance;
      (3) Supervision; and
      (4) Victoria class submarine.

   b. Denied: nil.

   c. Environmental: any conditions, day or night, at sea or alongside.

3. **Standard**: The VCS Senior Engineering Watchkeeper shall Manage Compliance with the SUBSAFE Policy (AFT0240) on a Victoria Class Submarine IAW references A8 – 10, A24 and A25 by:

   a. Acting as a Member of a Hazard Identification And Risk Assessment team; and

   b. Acting in support of the Submarine Safety Document Registration and Materiel Certificate.

4. **Training Limitations**: nil.

EO 005.01 – Act as a Member of a Hazard Identification and Risk Assessment

005.01.01 – Member of a Hazard Identification and Risk Assessment

Conditions		Sequence
Classroom		After PC 004
Time	**Classification**	**Instructor**
6 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturer – SUBSAFE

Learning Objectives	Method
Act as a Member of a Hazard Identification and Risk Assessment (HIRA) Team.	Interactive Lecture Practical Demonstration

Scenario	The submarine has encountered a defect which requires a HIRA. Utilizing their skills as an experienced OPCERT, the VCS SEWK is required to participate in the HIRA process.

Activity Steps	<u>Set up</u> (5 min)
	
	The instructor will ensure that the classroom is setup and ready to go. This includes a class with all the required reference and instructional material.
	
	<u>Introduction</u> (10 min)
	
	The instructor / Guest Lecturer will explain to the students the importance of the duties and responsibilities of the VCS SEWK while participating in the HIRA process.
	
	<u>Lecture</u> (200 min)
	
	Instructor shall discuss the importance of each of the learning objectives below. In particular, why the VCS SEWK must have a basic knowledge of the following SUBSAFE organizations, their purpose, the processes – in order to properly participate in the HIRA process:
	
	• SUBSAFE:
	  ○ Purpose;
	  ○ Policy; and
	  ○ Functions.
	
	• Submarine License:
	  ○ Purpose;
	  ○ Types;
	  ○ Levels;

- Building; and
- Pillars (certificates).

- Risk and Risk Management:
  - Definition;
  - Components;
  - Matrix;
  - Levels; and
  - Hazard.

- Conduct a Hazards – Risk – Risk Management Exercise;
- Review Materiel Certificate Documents;
- Risk Analysis and Risk Matrix;
- Demonstrate a SUBSAFE Exception Example; and
- Demonstrate the SUBSAFE Risk Management Summary Card (BINGO).

Demo (20 min)

The instructor will act as the VCS SEWK and utilizing one of the scenarios demonstrate what is expected of the VCS SEWK while participating in a HIRA, and why.

Scenario (50 min for class)

The instructor will have a series of scenarios available for use. A scenarios will be given to the class whom together, utilizing their basic knowledge of SUBSAFE organizations, purposes, and the processes, will conduct their own Hazards – Risk – Risk Management process and brief the instructor upon completion.

Review and Conclusion (15 min)

The instructor will review scenario areas of difficulty and ask the students if they have any questions.

Resources	References
1) Scenarios. 2) All reference material. 3) Guest Lecturer – SUBSAFE.	A8 – A10, A24 – A25

Notes to Instructor
Students should be familiar with each of the learning objectives by end of lectures.

EC 005.01 – Act as a Member of a Hazard Identification and Risk Assessment

Scope	Conditions	Sequence
EO 005.01	Classroom	After LG 005.01.01
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturer – SUBSAFE

Scenario	The submarine is at sea and about to embark on a critical mission. Within hours, the boat has lost use of the HP Bilge Pump and has been tasked by the Submarine Operating Authority to conduct a HIRA.
Assessment Steps	Instructions Given to Trainee. 1. EC 005 will be based on content from EO 005. 2. Each student will be assessed as the VCS SEWK, participating as a member of a Hazard Identification and Risk Assessment Team. 3. The instructor will have one scenario available for the entire class whom together, utilizing their basic knowledge of SUBSAFE organizations, purposes, and processes, will conduct their own Hazards – Risk – Risk Management process and brief the instructor upon completion. 4. The instructor will provide the scenario and all reference material listed below. 5. The instructor will act as the HIRA Team Chair. 6. Utilizing a scenario, the class (students) will be assessed using checklist EC 005.01 7. The must complete the EC in 40 min. 8. On completion the instructor will take 10 min to inform the class of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Scenarios 2) All reference material 3) Guest Lecturer – SUBSAFE	A8 – A10, A24 – A25

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

Item	EC 005.01 Act as a Member of a Hazard Identification and Risk Assessment	SAT	UNSAT	Remarks
	• SUBSAFE ○ Purpose ○ Policy ○ Functions	☐	☐	
	• Submarine License ○ Purpose ○ Types ○ Levels ○ Building ○ Pillars (certificates)	☐	☐	
	• Risk and Risk Management ○ Definition ○ Components ○ Matrix ○ Levels ○ Hazard	☐	☐	
	• Conduct a Hazards – Risk – Risk Management Exercise	☐	☐	

Score: /4

The class must achieve SAT on all factors to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 005.02 – Act in support of the Submarine Safety Document Registration and Materiel Certificate Process
005.02.01 – Submarine Safety Document Registration and Materiel Certificate

Conditions		Sequence
Classroom		After EC 005.01
Time	**Classification**	**Instructor**
6 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturer – FTA

Learning Objectives	Method
Acting in support of the Submarine Safety Document Registration (SSDR) and Materiel Certificate Process for Safety in Victoria Class Submarines.	Interactive Lecture Practical Demonstration

Scenario	The submarine is completing its Extended Docking Work Period (EDWP) and the SSDR is in the process of being re-issued.
Activity Steps	Set up (5 min) The instructor will ensure that the classroom is setup and ready to go. This includes a class with all the required reference and instructional material. Introduction (10 min) The instructor / Guest Lecturer will explain to the students the importance of the VCS SEWK acting in support of the Submarine Safety Document Registration and Materiel Certificate process. Lecture (250 min) Instructor shall discuss the importance of each of the learning objectives below. In particular, why the VCS SEWK must have both an understanding of basic SUBSAFE principals (LG 05.01.01), along with the following guidance and direction: • General Quality Management Policy for Submarines; • The Particular Approach for Victoria Class Submarines; • Quality Assurance Requirements for First Level Systems and equipment; • Procurement, Manufacture, Repair, installation and Stockholding requirements for in-service support; • Quality Records Systems – First Level Quality Assurance;

	• Use of items not fully Quality Assured in First Level Areas and use of Fully Assured Items in Non-First Level Application; and • Audits and Evaluations. <u>Guided Discussion</u> (30 min) After instruction, the instructor will ask students what the main takeaway was and if it changed their prospective on the importance the SSDR, and why. <u>Conclusion</u> (5 min) Ask if there are any questions. Reiterate the Submarine Safety Document Registration and Materiel Certificate process.

Resources	References
1) Scenarios. 2) All reference material. 3) Guest Lecturer – FTA.	A8 – A10, A24 – A25

Notes to Instructor
Students should be familiar with each of the learning objectives by end of lectures.

EC 005.02 – Act in support of the Submarine Safety Document Registration and Materiel Certificate Process

Scope	Conditions	Sequence
EO 005.02	Classroom	After LG 005.02.01
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	VCS SEWK Qualified Guest Lecturer – FTA

Scenario	The submarine is completing its Extended Docking Work Period (EDWP) and the SSDR is in the process of being re-issued.
Assessment Steps	Instructions Given to Trainee: 1. EC 005.02 will be a written test based on content from EO 005.02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. All References will be provided. 9. Pass mark is 70% 10. You will have 1 x 40 min period to complete this exam. 11. Do you have any questions before you start? 12. On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Classroom. 2) Exam. 3) All reference material. 4) DS Solution.	A8 – A10, A24 – A25

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

PC 005 – Manage Compliance with the SUBSAFE Policy on a Victoria Class Submarine as a SEWK

Scope	Conditions	Sequence
PO 005	Classroom	After EC 005.02
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	VCS SEWK Qualified

Scenario	The VCS SEWK is required to adhere to the SUBSAFE Policy both as a member of a HIRA, and at all times during the MAT Cert process.
Assessment Steps	Instructions Given to Trainee: 1. PC 005 will be a written test based on content from PO 005. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. All References will be provided. 9. Pass mark is 70% 10. You will have 1 x 40 min period to complete this exam. 11. Do you have any questions before you start? 12. On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Classroom 2) Exam 3) All reference material 4) DS Solution	A8 – A10, A24 – A25

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

PO 006 – Manage Submarine Stability as a SEWK

1. **Performance**: Manage Submarine Stability as a SEWK.

2. **Conditions**:

   a. Given:

      (1) References;
      (2) Assistance;
      (3) Supervision; and
      (4) Victoria Class submarine;

   b. Denied: nil;

   c. Environment: any conditions, day or night, at sea or alongside.

3. **Standard**: The VCS Senior Engineering Watchkeeper shall manage submarine stability both at sea and alongside (WT0020 and WT0070) IAW references A8 – A10, A24 and A25 by:

   a. Advising the Ship Control Officer of the Watch (SCOOW) on Submarine Stability at sea; and

   b. Advising the Trimming Officer on Submarine Stability while alongside.

4. **Training Limitations**: nil.

EO 006.01 – Advise SCOOW on Submarine Stability at sea as SEWK

006.01.01 – Submarine Stability at Sea

Conditions		**Sequence**
Classroom		After PC 005
Time	**Classification**	**Instructor**
6 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturer – NAV ARC

Learning Objectives	**Method**
Advise SCOOW on Submarine Stability at Sea.	Interactive Lecture Practical Demonstration

Scenario	The submarine is at sea and the VCS SEWK has a requirement to ballast and trim the submarine via the trim bilge and ballast system – utilizing various tanks and reporting to the SCOOW.
Activity Steps	<u>Set up</u> (5 min) The instructor will ensure that the classroom is setup and ready to go with all the required reference and instructional material. <u>Introduction</u> (20 min) The instructor / Guest Lecturer will explain to the students the importance of ballasting and trimming the submarine at sea via the trim bilge and ballast system – utilizing various tanks and reporting to the SCOOW. <u>Lecture</u> (240 min) Instructor shall discuss the importance of each of the learning objectives below. In particular, why the VCS SEWK must remain fully cognizant of the following items: • Water Tight Integrity; • Status of all Bulk Liquids to include Consumables; • Maneuvering Limitations Diagram; • Basic Submarine Static Stability; • Positive Buoyancy considerations; • Neutral Buoyancy considerations; • Negative Buoyancy considerations; and

	• Critical Hole Size. Guided Discussion (30 min) After instruction, the instructor will ask students what the main takeaway was and if it changed their prospective on the ballasting and trimming the submarine at sea, and why. Conclusion (5 min) Ask if there are any questions.

Resources	References
1) Guest Lecturer – NAV ARC. 2) All reference material.	A1 – A3, A6, A19, A22, A27, A31 – A32, A38, B10, B14, B19, B26

Notes to Instructor
Students should be familiar with each of the learning objectives by end of lectures.

EC 006.01 – Advise SCOOW on Submarine Stability at Sea as the SEWK

Scope	Conditions	Sequence
EO 006.01	Classroom	After LG 006.01.01
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	Guest Lecturer – NAV ARC

Scenario	The submarine is at sea and the VCS SEWK has a requirement to ballast and trim the submarine via the trim bilge and ballast system – utilizing various tanks and reporting to the SCOOW.
Assessment Steps	Instructions Given to Trainee: 1. EC 006.01 will be a written test based on content from EO 006.01. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. All References will be provided. 9. Pass mark is 70% 10. You will have 1 x 40 min period to complete this exam. 11. Do you have any questions before you start? 12. On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources 1) Classroom. 2) Exam. 3) All reference material. 4) DS Solution.	**References** A1 – A3, A6, A19, A22, A27, A31 – A32, A38, B10, B14, B19, B26
Notes to Instructor Nil	
Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

EO 006.02 – Advise Trimming Officer on Submarine Stability Alongside as a SEWK

006.02.01 – Submarine Stability Alongside

Conditions		Sequence
Classroom		After EC 006.01
Time	**Classification**	**Instructor**
6 X 50 Minute Periods	Unclassified	VCS SEWK Qualified Guest Lecturer – NAV ARC

Learning Objectives	Method
Advise Trimming Officer on Submarine Stability Alongside.	Interactive Lecture Practical Demonstration

Scenario	The submarine is alongside and the VCS SEWK is required to trim the submarine down for maintenance, or set the trim prior to sailing via the trim bilge and ballast system via various tanks and report to the Trimming Officer.
Activity Steps	Set up (5 min) The instructor will ensure that the classroom is setup and ready to go with all the required reference and instructional material. Introduction (20 min) The instructor / Guest Lecturer will explain to the students the importance of trimming the submarine down for maintenance, or setting the trim prior to sailing via the trim bilge and ballast system via various tanks, and reporting to the Trimming Officer. Lecture (240 min) Instructor shall discuss the importance of each of the learning objectives below IAW SOPs. In particular, why the VCS SEWK must remain fully cognizant of the following items: • Water Tight Integrity; • Status of all Bulk Liquids to include Consumables; • Basic Submarine Static Stability; • Positive Buoyancy considerations; and • Critical Hole Size.

	Guided Discussion (30 min) After instruction, the instructor will ask students what the main takeaway was and if it changed their prospective on trimming the submarine down for maintenance, or setting the trim prior to sailing, and why. Conclusion (5 min) Ask if there are any questions.

Resources	References
1) Guest Lecturer – NAV ARC. 2) All reference material.	A1 – A3, A6, A19, A22, A27, A31 – A32, A38, B10, B14, B19, B26

Notes to Instructor
Students should be familiar with each of the learning objectives by end of lectures.

EC 006.02 – Advise Trimming Officer on Submarine Stability Alongside

Scope	Conditions	Sequence
EO 006.02	Classroom	After LG 006.02.01
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	Guest Lecturer – NAV ARC

Scenario	The submarine is alongside and the VCS SEWK is required to trim the submarine down for maintenance, or set the trim prior to sailing via the trim bilge and ballast system via various tanks and report to the Trimming Officer.
Assessment Steps	Instructions Given to Trainee: 1. EC 006.02 will be a written test based on content from EO 006.02. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. All References will be provided. 9. Pass mark is 70% 10. You will have 1 x 40 min period to complete this exam. 11. Do you have any questions before you start? 12. On completion the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources 1) Classroom. 2) Exam. 3) All reference material. 4) DS Solution.	**References** A1 – A3, A6, A19, A22, A27, A31 – A32, A38, B10, B14, B19, B26
Notes to Instructor Nil	
Limitations	Nil
Impact of Failure	Instructor will review problem areas with the student, receive one supplementary EC and if unsuccessful, proceed with rest of the course.

PC 006 – Manage Submarine Stability as a SEWK

Scope	Conditions	Sequence
PO 006	Classroom	After EC 006.02
Time	**Classification**	**Instructor**
1 x 50 Minute Period	Unclassified	VCS SEWK Qualified

Scenario	The VCS SEWK is required to manage Submarine Stability both at sea and alongside.
Assessment Steps	Instructions Given to Trainee: 1. PC 006 will be a written test based on content from PO 006. 2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam. 3. Do not open the booklet until you are told to do so. 4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet. 5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s). 6. Read each question carefully, and all responses fully, before answering the question. 7. If you have any questions during the exam, raise your hand. 8. All References will be provided. 9. Pass mark is 70%. 10. You will have 1 x 40 min period to complete this exam. 11. Do you have any questions before you start? 12. On completion, the instructor will take 10 min to inform the student of a pass or fail, and on what aspects they did well and need to improve on.

Resources	References
1) Classroom. 2) Exam. 3) All reference material. 4) DS Solution.	A1 – A3, A6, A19, A22, A27, A31 – A32, A38, B10, B14, B19, B26

Notes to Instructor
Nil

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

CHAPTER 3: ASSESSMENT OF TRAINEE

GENERAL

1. All Performance Checks (PCs) are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessment and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Each EO must have at least one EC. EC results do not factor into a trainee's successful completion of the PO.

3. Performance Checks (PCs). PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be completed IAW the standards delineated in this QSP and IAW CFITES Volume 7 (Evaluation).

PROGRESS MONITORING

5. Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Volume 6 Conduct of Instruction Programmes.

6. Continuous monitoring of trainee progress is required to provide the following:

   a. Early warning of trainee difficulties;
   b. Feedback on the effectiveness of training;
   c. Information for the Training Review Board (TRB) (if required); and
   d. Information to assist in the completion of the CF 377 Course Report.

7. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment (TE) SOs or SOPs.

8. PC failures and supplemental PC failures are to be handled IAW TE SOs and SOPs. Supplemental ECs and PCs shall be given no less than 24 hours after the trainee being informed of their EC or PC failure.

TRAINING REVIEW BOARD

9. The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54 Training Review Board Process.

COURSE REPORTS

10. Trainees shall be graded pass or fail IAW NPTORD 4500-51.

11. Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

12. Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

13. A record of trainee counselling shall be maintained by the TE.

ANNEX A – MAIN REFERENCES

REF	TITLE	
A1	Victoria Class Submarine Standing Orders (VCSSOs)	
A2	Victoria Class Submarine Standard Operating Procedures (VCSOPs)	
A3	Victoria Class Submarine Emergency Operating Procedures (VCEOPs)	
A4	NAVORD 1002-0 – RCN Safety and Environmental Program Management	
A5	CSFORD 1006-1 – CSF Safety and Environment Program Management	
A6	Victoria Class Safety and Environmental Management System	
A7	SAMPLE – Fueling Orders	
A8	NAVORD 1150-0 Submarine Safety (SUBSAFE) Programme	
A9	SAMPLE – SUBSAFE Advisory	
A10	SAMPLE – SUBSAFE Directive	
A11	Guide to the Divisional System	
A12	NAVORD 1001-0 – Royal Canadian Navy Code of Conduct	
A13	NAVORD 1010-0 – RCN Defence Ethics Policy	
A14	NAVORD 1010-1 – RCN Defence Ethics Program	
A15	NAVORD 1020-1 – Naval Order – Hateful Conduct	
A16	CSFORD 4500-54 – Training Review Boards	
A17	CSFORD 5555-2 – Canadian Submarine Force Order – MOSID Champions	
A18	Captains Standing Orders	
A19	Captains Ships Book	
A20	Propulsion Power Statement	
A21	DRMIS Guide for Victoria Class Submarines	
A22	Naval Engineering Manual (NEMs)	
A23	Naval Material Management System (NaMMs)	
A24	Quality Assurance for Safety in Submarines	
A25	Submarine Material Management and Certification	
A26	SAMPLE – Docking and Undocking Plan	
A27	Manual of Trim and Stability	
A28	SAMPLE – FTA Trials Agenda	
A29	Tiered Readiness Program	
A30	SAMPLE – Night Order Book	
A31	SAMPLE – Engineering Officer Technical Instruction (EOTI)	
A32	SAMPLE – Engineering Officer Technical Memorandum (EOTM)	
A33	SAMPLE – Technical Bulletin	
A34	NAVORD 3250-7 Naval Operational Deficiency (NAVOPDEF) Messages	
A35	SAMPLE – NAVOPDEF Message	
A36	HQ1 Equipment to include: Submarine Incident Board TABs – BR 3623(01)(2)(3)(4) TWOPS Composite Drawings Electrical Systems	HQ1 Equipment to include: Change Over Switch Book Smoke Clearance Cards Liquid Chalk Markers Battery powered little sister Spare tapes

	Mechanical Systems Weapons Systems Command Briefing Cards Compartment Kill Cards	Sound powered Telephones PRC Radio DC Man Vol 4 Aide Memoire WHMIS MSDS Handbook
A37	NEMSOP – Flow Chart	
A38	FTA SSO Subs Standard Operating Procedures to include: FTA SUBS SOP 002 Designation of Materiel Certification FTA SUBS SOP 003 Guidelines for Materiel Certification Review FTA SUBS SOP 005 Understanding DRMIS FTA SUBS SOP 008 Short Work Period Planning and Execution FTA SUBS SOP 009 Submarine Work Period Management and Tracking FTA SUBS SOP 010 Useful References	
A39	NEMTI 001 NEMTI 002	
B1	BR 1966(2) – Firefighting Systems & Equipment's	
B2	BR 1966(3) – Escape and Salvage Systems and Equipment	
B3	BR 1966(4) – VPDC Systems & Equipment – Index	
B4	BR 1966(5) – Main Battery Systems & Equipment – Index	
B5	BR 1966(6) – 24 VDC Systems & Equipment – Index	
B6	BR 1966(7) – AC Systems & Equipment – Index	
B7	BR 1966(8) – Diesel Engines, Installation, Systems, & Equipment – Index	
B8	BR 1966(9) – Prop: Electrical Systems & Equipment – Index	
B9	BR 1966(10) – Prop: Mechanical Systems & Equipment – Index	
B10	BR 1966(11) – Compressed Air & Nitrogen Systems & Equipment – Index	
B11	BR 1966(12) – Ventilation, Air Conditioning & Snort Induction – Index	
B12	BR 1966(13) – Chilled Water Systems & Equipment – Index	
B13	BR 1966(14) – Ships Hydraulic Systems & Equipment – Index	
B14	BR 1966(15) – Trim, Bilge & Ballast Systems	
B15	BR 1966(16) – Ships SW System & Equipment	
B16	BR 1966(17) – Ships & Domestic FW System & Equipment	
B17	BR 1966(18) – Desalination Plant	
B18	BR 1966(20) – Surveillance System and Equipment	
B19	BR 1966(21) – Control Consoles – Index	
B20	BR 1966(22) – Rudder & Hydroplanes, Control System & Equip – Index	
B21	BR 1966(23) – Masts & Associated Equipment – Index	
B22	BR 1966(24) – Ship's Stores Refrigeration	
B23	BR 1966(25) – Domestic Equipment	
B24	BR 1966(30) – Lighting Systems & Equipment – Index	
B25	BR 2075(K) – Submarine Main Batteries	
B26	BR 3623(01) – TAB's – Ships Systems	
B27	BR 3623(02) – TAB's – Propulsion	
B28	BR 3623(03) – TAB's – Tactical Weapons Systems	
B29	BR 3623(04) – TAB's – Electrical Systems	
B30	BR 4500(526) – Static Frequency Changer, MK38SFC	
B31	BR 4500(529) – Static Frequency Changer, MK36SFC	

B32	BR 4500(701) – Batteries, Primary & Secondary
B33	BR 4569 – Pressure & Differential Pressure Transducers & Associated Equip
B34	BR 6528(301) – Paxman Valenta Diesel Engine 16 RPA 200z
B35	BR 6528(303) – Valenta Engines Dockyard Overhaul/Repair
B36	BR 8223 – Breathing Apparatus

ANNEX B – TASK LIST

This table provides an overview of all tasks considered and where they are trained within the QSP if selected. No train substantiations are in the Record of Decisions annex.

Task #	Task Statement	T/NT	PO
AFT0240	Monitor Compliance with Policy	T	005
AFT0400	Report maintenance activities within defence resource management system (DRMIS)	T	003
AFT0450	Review engineering change proposal	T	003
AFT0500	Review Pre Installation Failure (PIF) reports	T	003
AFT0860	Advise on unsatisfactory condition reports (UCR)	T	003
AFT0870	Analyze equipment health monitoring data	T	003
CT0010	Advise command on marine systems engineering parameters	T	001
CT0020	Advise command on marine systems operational capabilities and limitations	T	001
CT0030	Advise command on power generation and distribution systems	T	001
CT0040	Advise command on propulsion capabilities and limitations	T	001
IT0020	Advise command on damage control actions	T	002
IT0400	Advise command on damage control system status	T	002
OT0130	Manage maritime vessel fuelling (internal fuel / providing / consolidation)	T	004
OT0190	Supervise fuel quality testing	T	004
PT0010	Advise command on ship's power configuration	T	001
PT0020	Advise command on ship's power isolation	T	001
WT0020	Advise command on ship's stability	T	006
WT0070	Manage ship's stability	T	006
Added Tasks			

ANNEX C – RECORD OF DECISIONS

Dates: 12 January – 22 January 2021

Location: Classroom ATC-2, Building N92-A, Naden

Working Group Members:

Chair – Lt(N) Bryson, CSF Sustainment

Secretary – S1 Corner, HMCS Victoria

TDO – Lt(N) Guthrie, NTDC(A) Subs

SME – CPO2 Horobin, Sea Training (Submarines) Chief Engineer

SME – CPO2 Niezen, Sea Training (Submarines) Chief Electrician

SME – PO1 Hamilton, NFS(A) Submarine Technical Training Manager

SME – PO1 Langley, NFS(P) Shops I/C

SME – PO2 Puxty, Electrical Maintenance Supervisor HMCS Chicoutimi

Guests – on a case by case basis to be captured in the Record of Proceedings

Working Group Objective:

1. Context provided to the Working Group was: If you recall, in 2017 the Mar Tech trade was officially stood up, launching a new era in the operation and maintenance of highly technical propulsion, mechanical and power generation systems, enabling RCN warships and submarines to float, move and fight. This new trade also moved away from the certification process. Namely, the Cert 2 became the Engineering Watchkeeper (EWK) – our version of the Op Cert, the Cert 3 was replaced by SEWK (why we are convening), and the Cert 4 was removed in its entirety. As this new trade evolved and worked through legacy coursing, the submarine force, through the remaining Cert 3 and Cert 4 members was able to retain competency, knowledge, and the required experience to support maintenance and technical authorities, as well as operations at sea. More importantly, these members were able to provide timely mentorship to submariners transitioning to the new trade. However, the sheer number of recent retirements, coupled with the subsequent loss of corporate knowledge has resulted in members (through no fault of their own) fulfilling roles they haven’t been fully trained for (current MSECs of both HMCS Victoria and Windsor).

2. Given all that in para 1, the objective of this Working Group is to:

- Identify the delta created by the removal of the Cert 3 and Cert 4

- ensure that all future SEWKs and MSECs have the right skill set in order to best support both Command and the MSEO

- determine prerequisite rank, coursing, OJPR, or EQ consensus on RQ-PO2 Course USQ Mech

- discuss the 500 hours as a qualified EWK

- determine the impact on other reference material such as VCSSO, Submarine Licensing etc.

- upon completing the course and OJPR, should there be a formal Board – similar to the legacy Cert 3 and 4

3. Specification Tasks:

Task #	Task Statement	Substantiation
AFT0240	Monitor Compliance with policy	Required in the performance of SEWK duties and responsibilities
AFT0400	Report maintenance activities within Defence Resource Management Information System (DRMIS)	Required in the performance of SEWK duties and responsibilities
AFT0450	Review Engineering Change (EC) proposal	Required in the performance of SEWK duties and responsibilities
AFT0500	Manage Pre Installation Failure (PIF) reports	Required in the performance of SEWK duties and responsibilities
AFT0860	Manage Unsatisfactory Condition Reports (UCR)	Required in the performance of SEWK duties and responsibilities
AFT0870	Analyze equipment health monitoring data	Required in the performance of SEWK duties and responsibilities
CT0010	Advise Command on Marine Systems Engineering parameters	Required in the performance of SEWK duties and responsibilities
CT0020	Advise Command on Marine Systems Operational capabilities and limitations	Required in the performance of SEWK duties and responsibilities
CT0030	Advise Command on Power Generation and Distribution Systems	Required in the performance of SEWK duties and responsibilities
CT0040	Advise Command on Propulsion Capabilities and Limitations	Required in the performance of SEWK duties and responsibilities
IT0020	Advise Command on Damage Control actions	Required in the performance of SEWK duties and responsibilities
IT0040	Advise Command on Damage Control system status	Required in the performance of SEWK duties and responsibilities

OT0130	Manage Maritime Vessel Fuelling (internal fuel / providing / consolidation)	Required in the performance of SEWK duties and responsibilities
OT0190	Supervise Fuel Quality Testing	Required in the performance of SEWK duties and responsibilities
PT0010	Advise Command on Ship's Power Configuration	Required in the performance of SEWK duties and responsibilities
PT0020	Advise Command on Ship's Power Isolation	Required in the performance of SEWK duties and responsibilities
WT0020	Advise Command on Ship's Stability	Required in the performance of SEWK duties and responsibilities
WT0070	Manage Ship's Stability	Required in the performance of SEWK duties and responsibilities

4. Discussions / Decisions:

Day One – Tuesday 12 January

Introduction

The Board commenced with the Chair welcoming all members, giving an introduction, and asking each of the members to provide a quick brief on their background and any related QSP experience. The Chair then went on to brief the Board on the rational for convening the Senior Engineering Watchkeeper Board (captured above).

The TDO, LT(N) Guthrie provided an excellent overview of the QSP process. He described the various quality control phases of CFITES while at the same time providing an example of Performance Orientated Training and its various components. He stated that as the Board progressed, there would be a reiteration of the various phases of CFITES. Prior to beginning this important task, the Chair asked if there were any questions. With none, the Board commenced its work.

The Board then went onto discuss the Task List above (de-scoped from the MAR TECH JTAR) and determined that all of the tasks are to be trained. One task – Submarine Certification Licensing Management System – SUBSAFE (not captured in the JTAR) was added to the Task List. Subject task was discussed with the Mar Tech Training Manager and determined that this would fall under Task AFT 0240 Monitor Compliance with policy.

The Board settled on a maximum course load of six, minimum two, and optimum of four.

The Board also discussed the course prerequisites, especially, the USQ Mech, as it was essential that both streams attend the course in order to acquire the right skill sets to perform the duties and responsibilities of a VCS SEWK, and ultimately, an MSEC.

Day Two – Wednesday 13 January

The Chair began by recapping the previous day's events stating now that train tasks had been determined – it was time to develop the Enabling and Performance Objectives. For simplicity, the Board determined that those tasks considered to be an enhancement of the already acquired OPCERT skill sets would be tackled first. As a result, the Board captured those tasks within the following PO's and EO's:

PO 001	Advise OOW on Submarine Marine Systems Engineering as the SEWK
EO 001.01	Advise OOW on Submarine Power Configuration
EO 001.02	Advise OOW on Submarine Propulsion Capabilities and Limitations
EO 001.03	Advise OOW on Submarine Mechanical Systems

PO 003	Supervise Victoria Class Submarine Maintenance Activities as a SEWK
EO 003.01	Respond to Engineering Change Proposals
EO 003.02	Respond to Equipment Health Monitoring Data
EO 003.03	Supervise Defence Resource Management Information System
EO 003.04	Manage Marine Systems Engineering Reports

Day Three – Thursday 14 January and Day Four – Friday 15 January

Deliberations commenced with Board members discussing how to tackle Submarine Certification Licensing Management System (SUBSAFE) – the one task missing from the JTAR. After revisiting the train decision stance previously taken by the Board, all were still adamant that SUBSAFE be included in the curriculum as there will undoubtedly be a requirement for the VCS SEWK to participate in both a HIRA and Material Certification process. Because of the complexities of SUBSAFE, Board members felt that an SME from CSF Readiness is best suited to instruct HIRA procedures, while an SME from FTA is best suited to instruct Material Certification. With that, the following were captured in the two chunks:

EO 005.01	Act as a Member of a Hazard Identification And Risk Assessment
EO 005.02	Act in support of the Submarine Safety Document Registration and Materiel Certificate Process

In keeping with subject matter complexities, the Board also felt that a NAV ARC is ideally suited to instruct the following two submarine stability EC's:

EO 006.01	Advise SCOOW on Submarine Stability at Sea as the SEWK
EO 006.02	Advise Trimming Officer on Submarine Stability Alongside

The Board also felt very strongly that both the EC's and PC should be solely based upon procedures, vice practical calculations or exercises. After consultation with the Mar Tech Training Manager, it was determined that the proposed task would be covered with the existing AFT0240 Monitor Compliance with policy, task.

Day Five – Monday 16 January

Unlike the Surface Ship Counterpart – VCS SEWK is completely embedded the HQ1 organization and highly relied upon by the Incident Controller as the person with right skill sets and platform situational awareness, they provide technical expertise in support of the Incident Controllers recommendations to Command. Utilizing the Damage Control JTAR tasks, the following EO's were developed:

EO 002.01	Advise Incident Controller on Immediate Damage Control Actions
EO 002.02	Advise Incident Controller on Power and System Isolations
EO 002.03	Provide Recommendations to the Incident Controller
EO 002.04	Manage SEWK Transition From HQ1 to Either HQ2 or HQ3

Day Six – Tuesday 17 January

As the VCS SEWK plays a key role in the fueling and de-fueling process, the following EO's were developed:

EO 004.01	Verify Fueling and Defueling System Line Up
EO 004.02	Conduct Fueling and Defueling Safety Brief
EO 004.03	Secure from Fueling and Defueling

Before concluding for the day, Board members felt very strongly that prior to awarding the VCS SEWK Qual, there be an be OJPR to further refine those skill sets linked directly to the following key advisory or supervisory roles:

001 – Advise OOW on Submarine Marine Engineering Systems as the SEWK
002 – Advise Incident Controller as a SEWK
003 – Supervise Victoria Class Submarine Maintenance Activities as a SEWK
004 – Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK

Moreover, Board members felt that once the OJPR was completed, there be a requirement for trainees to successfully challenge an oral board with the Engineering Officer as the Chair – along with the MSEC, PO1 Mar Tech Mech / PO1 Mar Tech Elec, or their designated representatives.

Day Seven – Wednesday 18 January

The Board spend much of the morning discussing skill sets, deck plate experience, and the institutional leadership nexus associated with the VCS SEWK. Although the Board managed to capture all of the tasks associated with the JTAR (including SUBSAFE) within EO's, there was concern that some aspects of institutional – supervisory and managerial leadership skill sets were not appropriate for some of the VCS SEWK tasks – but more fitted for the MSEC.

Given the above, through consultation with the Occupational Manager, the Board decided that it would leverage off the existing Halifax Class MSEC EQ Package. Those supervisory and managerial leadership skill sets are now captured within the "new" VCS MSEC EQ Package. In doing so, this will now provide Mar Tech Managers the required experience to be able to fulfill the position of MSE Department Coordinator (MSEC) on board HMC Submarines.

Day Eight – Thursday 19 January

Due to unforeseen circumstances – two key SME's were required to depart the Board on short notice to support submarine operational requirements. In view of this, and the fact that these two SME's were absolutely integral to the success of the Board, the Chair paused the procedures – declaring a hiatus until they (SME's) were available on a non-interference basis.

Friday 21 May

After many weeks of the QSP Chair and SME's convening on both an availability and as required basis, including dialog with NTDC (A) and the Occupation Manager, along with advice from TDO, the Board felt comfortable submitting this QSP to NTDC(A).

5. Guidance to Development and/or Instructional Staff:

Both of these packages align with the HFX Class packages, are not linked to trainers or a hull, and can therefore be instructed on either coast (including guest lecturers and tours).

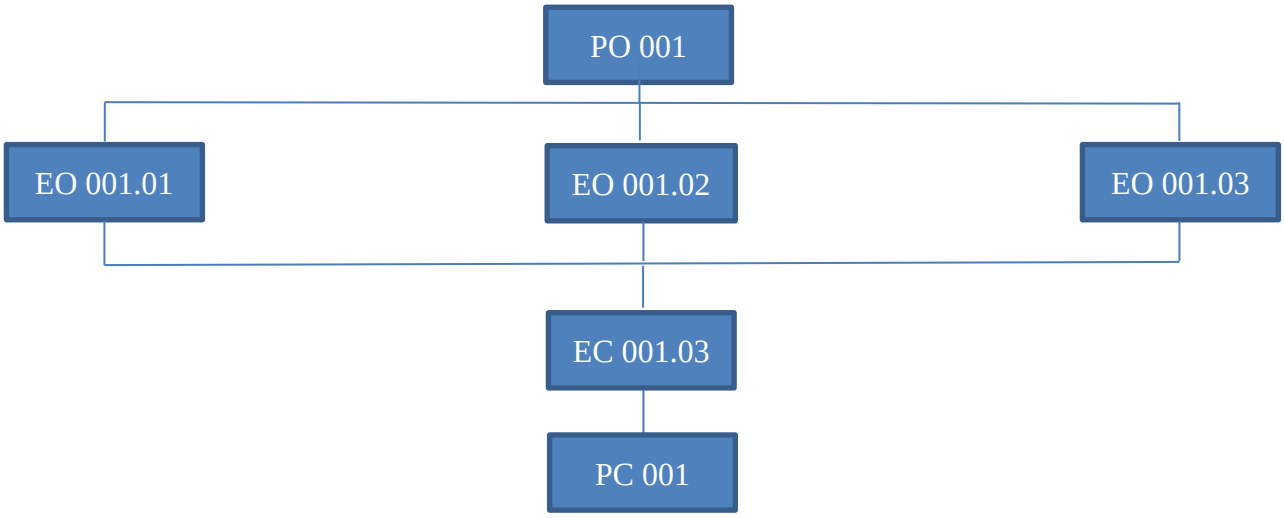
6. Follow Up Action:

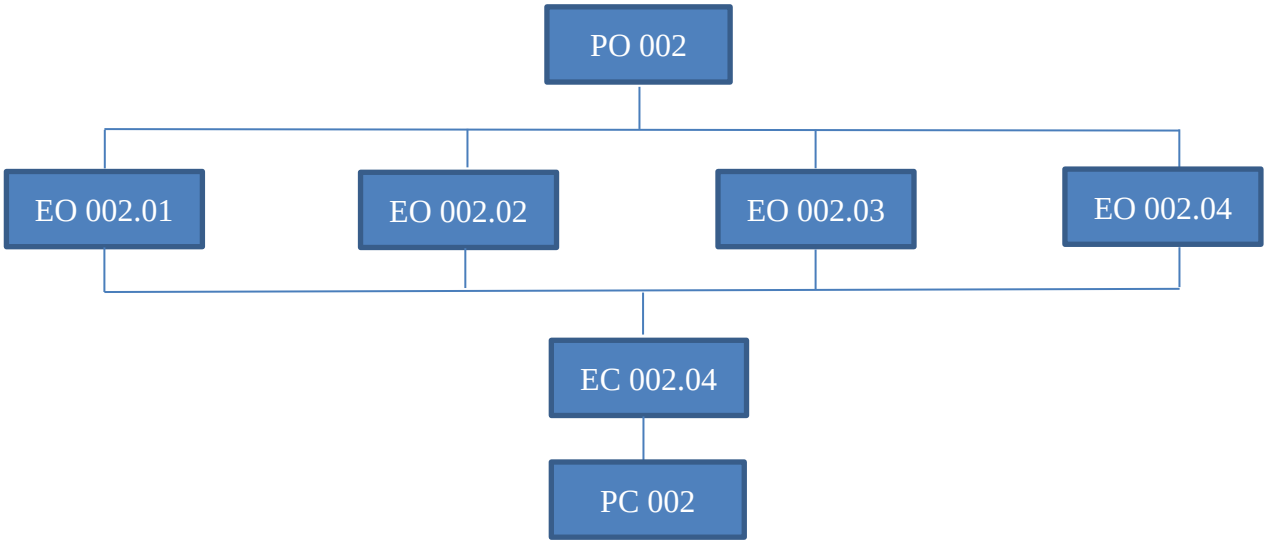
Item	Action	OPI	Expected Date of Completion
1	TDO review	NTDC (A)	Complete
2	Develop the appropriate: EO scenarios and quizzes, EC and PC scenarios and both written and practical exams.	NFS (A)	TBD
3	Requirement for both a MITE Code and NQual.	NTDC (A)	TBD
4	Arrange piloting for either east or west coast.	NTDC (A) / NFS (A)	TBD
5	Many of the EO's require participation of Guest Lecturers, and the arrangement of tours that may require advance notice.	NFS (A)	TBD

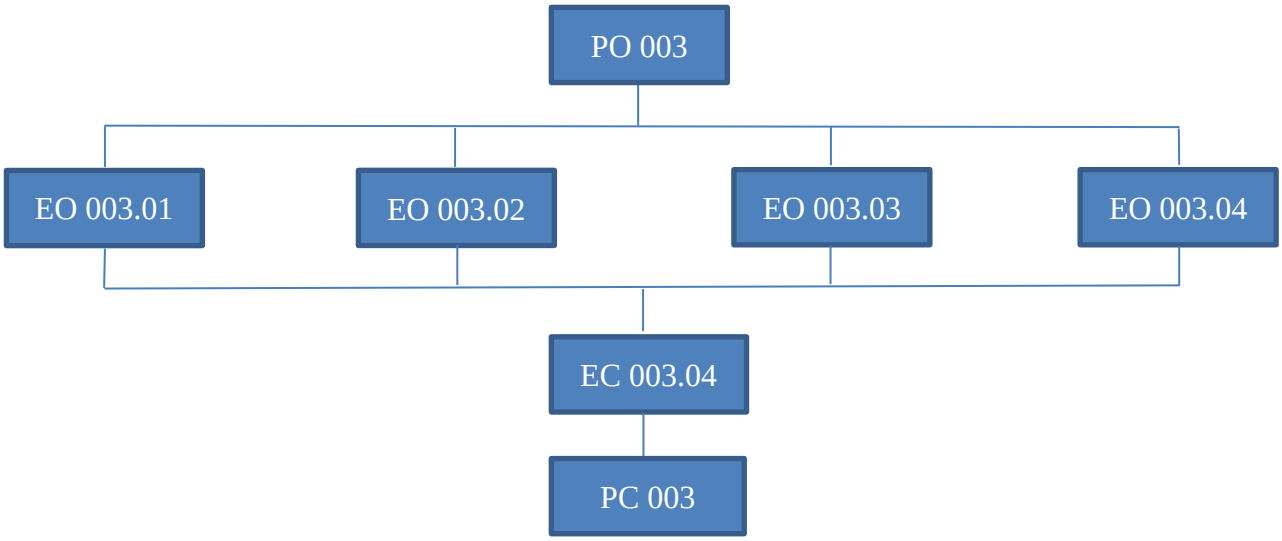
7. Course History:

Nil – Captured in the Working Group Objective.

ANNEX D – QSP SCALARS



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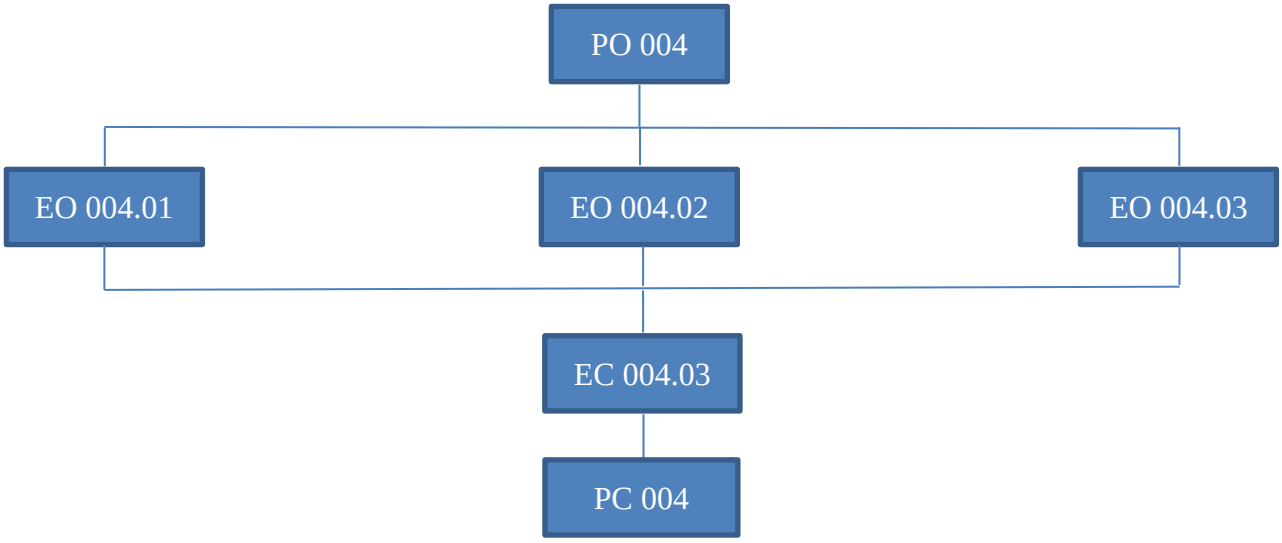
EO 003.02
EO 003.03
EO 003.04
EC 003.04
PC 003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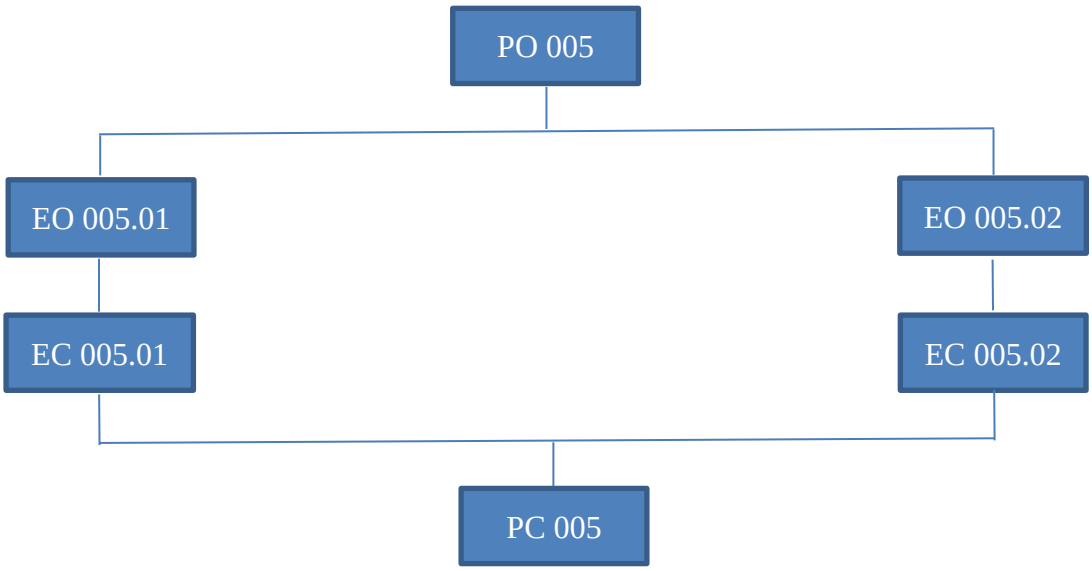
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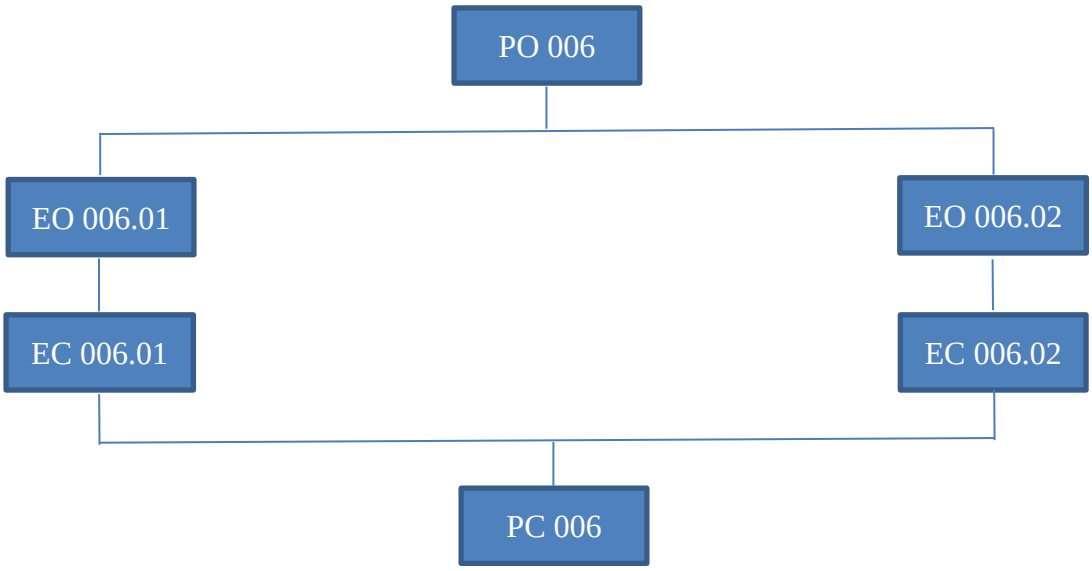
EC 005.01
EC 005.02
PC 005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EO 006.02
EC 006.01
EC 006.02
PC 006

ANNEX E – COSTING INFORMATION

Include information about cost in whichever format is applicable (**Not to be deleted**).
Estimates only, based on CF Cost Factors Manual
****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: SAC D		Serial:				
TD	No. Of Trainees / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	6	$1,000.0	N/A	$8,000	$8000	$40,000.00
R&Q	6	$36.0	102	$29,376	$29,376	$146,880.00
Incidentals	6	$17.5	102	$14,280	$14,280	$71,400.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	102	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$336,205**

ANNEX F – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.03	EO 001.01 to EO 001.03	To assess trainee's competence on Advising OOW on Submarine Marine Engineering Systems as the SEWK	Practical	Trainee must obtain SATs IAW EC 001.03 Checklist
PC 001	PO 001	To assess trainee's competence on Advising OOW on Submarine Marine Engineering Systems as the SEWK	Practical	Trainee must obtain SATs IAW PC 001 Checklist
EC 002.04	EO 002.01 to EO 002.04	Advise Incident Controller as a SEWK	Practical	Trainee must obtain SATs IAW EC 002.04 Checklist
PC 002	PO 002	Advise Incident Controller as a SEWK	Practical	Trainee must obtain SATs IAW PC 002 Checklist
EC 003.04	EO 003.01 to EO 003.04	Supervise Victoria Class Submarine Maintenance Activities as a SEWK	Written	Trainee must obtain Pass mark of 70%
PC 003	PO 003	Supervise Victoria Class Submarine Maintenance Activities as a SEWK	Written	Trainee must obtain Pass mark of 70%
EC 004	EO 004.01 to EO 004.03	Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK	Practical	Trainee must obtain SATs IAW EC 004.03 Checklist
PC 004	PO 004	Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK	Practical	Trainee must obtain SATs IAW PC 004 Checklist
EC 005.01	EO 005.01	Act as a Member of a Hazard Identification And Risk Assessment	Practical	Class / Trainee must obtain SATs IAW EC 005 Checklist
EC 005.02	EO 005.02	Act in support of the Submarine Safety Document Registration and Materiel Certificate Process	Written	Trainee must obtain Pass mark of 70%

PC 005	PO 005	Manage Compliance with the SUBSAFE Policy on a Victoria Class Submarine as a SEWK	Written	Trainee must obtain Pass mark of 70%
EC 006.01	EC 006.01	Advise SCOOW on Submarine Stability at Sea as the SEWK	Written	Trainee must obtain Pass mark of 70%
EC 006.02	EC 006.02	Advise Trimming Officer on Submarine Stability Alongside	Written	Trainee must obtain Pass mark of 70%
PC 006	PC 006	Manage Submarine Stability as a SEWK	Written	Trainee must obtain Pass mark of 70%

ANNEX G – TEL DEVELOPMENT OPPORTUNITIES

Title	Type	Description of Requirement	Lesson	LSR #
Fueling System	Training Aid	Mock-up of VCS Fueling System to be used to teach fueling and defueling. Could be 3D printed diagram with indication of open and shut valves.	LG 004.01.01 – 004.03.01, EC 004.03 and PC 004	TBD

ANNEX H – SCHEDULING INFORMATION

The recommended schedule is based on maximum load. Scheduling cells are to increase/decrease course length based on number of trainees enrolled on the course.

Scheduling Information													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL	Class	WB	Desk	Type	Qualification	Sec Cl
Residential Phase													
001.01.01	100	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
001.02.01	100	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
001.03.01	100	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
EC 001.03	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
PC 001	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
002.01.01	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
002.02.01	200	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
002.03.01	100	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
002.04.01	100	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
EC 002.04	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
PC 002	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
003.01.01	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
003.02.01	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
003.03.01	350	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
003.04.01	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
EC 003.04	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
PC 003	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
004.01.01	100	Classroom	X			X	X	X	X	6	Mil	VCS SEWK	UNC
004.02.01	50	Classroom	X			X	X	X	X	6	Mil	VCS SEWK	UNC
004.03.01	50	Classroom	X			X	X	X	X	6	Mil	VCS SEWK	UNC
EC 004.03	300	Classroom	X			X	X	X	X	6	Mil	VCS SEWK	UNC
PC 004	300	Classroom	X			X	X	X	X	6	Mil	VCS SEWK	UNC
005.01.01	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC

Scheduling Information													
Lesson		Location	IT Assets					Footprint			Instructor		
#	Time	Location	DWAN	CSNI	SSA	Proj	TEL	Class	WB	Desk	Type	Qualification	Sec Cl
EC 005.01	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
005.02.01	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
EC 005.02	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
PC 005	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
006.01.01	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
EC 006.01	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
006.02.01	300	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
EC 006.02	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC
PC 006	50	Classroom	X			X		X	X	6	Mil	VCS SEWK	UNC

ANNEX I – ON JOB PERFORMANCE RECORD

_____

Name and Initials Rank SN

_____

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IAW THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. The trainee will have 6 months from the date training commenced to complete this OJPR. Any extensions may be requested by the trainee's supervisor or divisional officer IAW NPTORD 4500-55 On-the-Job Training Management.

3. As the trainee completes each Performance Objective (PO) to the required standard, the Training PO will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

4. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a submarine, is not on issue to the unit, or is otherwise unavailable. If the training PO encounters any of these issues they are directed to contact NTDC (A) Sub Div. for guidance.

5. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGESS REVIEW

6. Progress review shall be carried out quarterly by the Divisional Officer and Section Head,

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

On-Job Training Performance Record				
PC	Title	Date	Signature	Comments
USQ Senior Engineering Watchkeeper VIC				
001	Advise OOW on Submarine Marine Engineering Systems as the SEWK			
002	Advise Incident Controller as a SEWK			
003	Supervise Victoria Class Submarine Maintenance Activities as a SEWK			
004	Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK			

7. The final portion of this OJPR is to successfully challenge an oral board. The Engineering Officer shall chair this board. At a minimum the other members will be MSEC, PO1 Mar Tech Mech / PO1 Mar Tech Elec or designated representatives.

TRAINING COMPLETION CERTIFICATE

8. _____ has completed the requirements from the OJPR and shall be awarded the qualification for USQ Senior Engineering Watchkeeper VIC (NQUAL), MITE Code 112796.

_____

Board Chair Date

_____

MSEC signature Date

_____

PO1 MAR Tech Signature Date

9. I certify that _____ has completed training to the standards stated in QSP USQ Senior Engineering Watchkeeper VIC.

_____

Commanding Officer Date

ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in this annex shall be used for the following information only:

    a. The unit which the training and assessment was carried out; and
    b. Any unusual circumstances which preclude compliance to an objective.

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
PO 001 Advise OOW on Submarine Marine Engineering Systems as a SEWK				
EO 001.01 Advise OOW on Submarine Power Configuration				
001.01.001	Conduct single MG operations			
EO 001.02 Advise OOW on Submarine Propulsion Capabilities and Limitations				
001.02.001	Conduct single armature operations.			
001.02.002	Identify links for breaker and contact bypass.			
001.02.003	Describe procedure for installing links.			
EO 001.03 Advise OOW on Submarine Mechanical Systems				
Demonstrate Procedure for Isolating Main High Pressure Air System				
001.03.001	State the step-by-step process.			
001.03.002	Point out all controls, valves, manipulated during a high pressure air failure.			
001.03.003	Diagnose the defect.			
001.03.004	Relate the system affected.			
001.03.005	Describe alternate system line-up, including lock out/tag out while the repairs are being affected.			
001.03.006	Describe the repair procedure to be used.			
Demonstrate Procedure for Isolating the Hydraulic System				
001.03.007	State step-by-step procedure			

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
001.03.008	Point out controls, valves, etc. that are manipulated during a hydraulic failure.			
001.03.009	Diagnose the defect.			
001.03.010	Relate the systems affected.			
001.03.011	Prescribe alternate system line up, including lock out/tag out, while repairs are carried out.			
001.03.012	Prescribe the repair procedure to be used.			
Prepare a Submarine Tank for Inspection				
001.03.013	Prepare a tank for inspection IAW NEMS			
Supervise Vacuum Test of a Submarine				
001.03.014	Inspect all Inner and Outer Boundaries			
001.03.015	Submarine correctly shut down.			
001.03.016	Only authorized personnel aboard and correctly positioned.			
001.03.017	Appropriate vacuum attained.			
001.03.018	Time limits not exceeded			
001.03.019	Submarine reverted to normal condition and equalized IAW SOPs once test completed.			
001.03.020	Assist EO to complete vacuum test report.			
PO 002 Advise Incident Controller as a SEWK				
EO 002.01 Act as Member of HQ1				
002.01.001	Act as SME during DC Evolution (min 3 times) 1.			

Trainee Rank: _____ Name: _____

PO/EO	PERFORMANCE STATEMENT	SIGNATURE	DATE	REMARKS (as required)
	2. 3.			
PO 003 Supervise Victoria Class Submarine Maintenance Activities as a SEWK				
EO 003.01 Supervise Victoria Class Submarine Maintenance Activities				
003.01.001	Complete a LOG REQ.			
003.01.002	Complete a TRAN REQ.			
003.01.003	Complete an EC.			
003.01.004	Complete a PIF.			
003.01.005	DIR OQE to an N1/ND			
PO 004 Supervise Fueling and Defueling on a Victoria Class Submarine as a SEWK				
EO 004.01 Supervise Fueling and Defueling on a Victoria Class Submarine				
004.01.001	Safety of personnel and submarine.			
004.01.002	Communications.			
004.01.003	System line up.			
004.01.004	Environmental precautions followed.			
004.01.005	Technical instructions and safety precautions observed.			
004.01.006	Change of heel does not exceed +/- 2 degrees.			
004.01.006	Systems correctly secured upon completion of fuelling/defueling			
004.03.007	Assist the fuelling I/C during a fuelling			
004.03.008	Act as the fuelling I/C during a fuelling			

APPENDIX 1

SENIOR ENGINEERING WATCHKEEPER

QUALIFICATION BOARD ASSESSMENT GUIDE

ORAL EXAMINATION BOARD PROCEDURE

1. The Oral Examination Board conducted at the end of the Senior Engineering Watchkeeper training package will measure the candidate's understanding of the material studied and the engineering systems they have learned to this point. In order to pass the board, the candidate must receive a mark of at least 60% with the first two questions also having a Pass/Fail criteria for the board (i.e. a failure of either results in a failure of the board). Board length will be set at a maximum of two (2) hours. Board dates and times will be coordinated by the submarines MSEO. The board shall be composed of the Submarine's MSEO (as board Chair), the Chief of the MSE Department and a PO1 MAR TECH of the opposite specialty as the MSEC

2. During the board the candidate shall not have access to any references, and will be supplied with the drawing (supplied by NTDC(A)). The board shall consist of five (5) questions. The first question will be a description of the system drawing. The candidate will be informed which diagram they are required to present 30 minutes prior to the board to allow the candidate to prepare. The second question will be based on a Leadership/ Administration scenario. The board will then take a break to determine if the candidate has successfully challenged the first two questions prior to carrying on with the rest of the board. The remaining questions shall consist of a stability question, a watchkeeping question, and a question based on a VCEOP with a follow on engineering problem scenario. The questions will be prepared from the question bank provided by NTDC(A).

3. In the event of a board failure or training difficulties, refer to the MAR TECH Senior Engineering Watchkeeper QSP.

4. Board Format (suggested timings for questions noted below)

Q1: Engineering drawing – describe a provided sketch (P/F)
Q2: Leadership/Administration (P/F)
Q3: Stability and liquid handling
Q4: General Technical knowledge, in relation to watchkeeping

Q5: VCEOP with a follow on based engineering problem (candidate will have time to put plan together) (P/F)

1. ENGINEERING DRAWING (drawing supplied) (45 min)

The candidate is provided with a drawing 30 minutes before the Board. The drawing is of a system with parameters/other details removed. The drawings to be used for this question are the following:

a. Electrical Power Generation and Distribution;

b. Propulsion/Shaftline;

c. Foreplanes;

d. Steering/Aft Planes; and

e. Fuel Oil Services.

The examination board members will have a copy of the drawing with all relevant information provided. The candidate must explain the following to the Examining Board:

a. System Description to include:

(1) Overview of system; and

(2) Purpose of system.

b. System Operation to include:

(1) Normal seagoing configuration; and

(2) Special operating states.

c. Component Description:

(1) Brief purpose and function of each component from the perspective of the Senior Engineering Watchkeeper.

d. Parameters

(1) Normal operating ranges; and

(2) Max/min values and warning/alarm points.

e. Direction of flow for:

(1) Fluids; and

(2) Energy, rotation or other mechanisms, as appropriate.

f. Abnormal operation:

(1) Alternate modes in response to a failure or breakdown and their limitations; and

(2) Considerations to the engineering plant in response to any abnormal mode.

g. Emergency Operation

2. LEADERSHIP/ADMINISTRATION (15 min)

The candidate will be given a Leadership/Administration scenario. The candidate is to provide actions and recommendation to the board. They will discuss relevant references, administrative requirements, options, courses of action, and recommendations to chain of command. The candidate should state relevant outside resources available to assist.

3. STABILITY AND LIQUID HANDLING (15 min)

The candidate will be given a stability question (ie. operation within the MLD envelope, critical hole size at depth). This question is intended to ensure the senior watchkeeper is faced with potential changes to stability and that the submarine is kept safe. Follow on aspects can include providing general recommendations to OOW and briefing MSEC/MSEO.

4. GENERAL TECHNICAL KNOWLEDGE (15 min)

The candidate will be given an abnormal watchkeeping question (e.g. failure of DEBHUV to open, single MG operations, etc), to identify the actions the senior watchkeeper should take. This question is intended to determine the watchkeeper's knowledge. This may include general recommendations to the OOW and briefing MSEC/MSEO.

5. SCENARIO BASED (30 min)

The candidate will be given a VCEOP scenario and will be asked to explain how they would react. Reactions to the scenario are to include:

a. watchkeeper actions;

b. listing personnel/technical resources available to assist;

c. HQ1 recommendations; and

d. briefs to MSEC/MSEO.

The candidate will react to the scenario as a watchkeeper initially. They will then be given time to compile their responses to follow on questions prior to responding to the board members.

Copy _____ of _____		
Senior Engineering Watchkeeper – Qualification Board Questions		
Candidate:	Submarine:	Date:
MSEO:	MSEC:	PO1 MAR TECH

INSTRUCTIONS

Questions are scored out of 10 pts; 0-8 pts for knowledge / understanding and 0-2 pts for presentation / conduct. A minimum of 60% is required for each of the first two questions in order for the board to carry on past the second question.

KNOWLEDGE / UNDERSTANDING:

0 No knowledge. Incapable of answering main part of the question, including where the system/equipment is located, what it looks like or how to begin even researching the topic.

1 Extremely low knowledge. Incapable of answering main part of the question other than where the system/equipment is located, what it looks like or knows where/who to go to in order to obtain more information.

2 Very low knowledge. Continuous assistance was required to answer the entire main question and answers were insufficient in detail.

3 Low knowledge. Continuous assistance was required to answer half or fewer of the main question and answers were insufficient in detail.

4 Marginal knowledge. Frequent assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

5 Adequate knowledge. Minimal assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

6 Proficient knowledge. Minimal assistance was required to answer the main question and follow up questions in acceptable detail.

7 Strong knowledge. No assistance was required to answer the main question and follow-up questions in acceptable detail.

8 Exemplary knowledge. Answered main question completely and elaborated in detail on specific aspects such that follow-up questions were unnecessary.

PRESENTATION / CONDUCT:

0 Displayed no confidence or professionalism with board members. Answers/diagrams were vague, unorganized and non-committal.

0.5 Displayed poor confidence or professionalism with board members. Answers/diagrams were partially vague and partially unorganized.

1	Displayed confidence and professionalism with board members. Answers/diagrams were organized.
1.5	Displayed strong confidence and professionalism with board members. Answers/diagrams were organized well.
2	Displayed exemplary confidence and professionalism with board members. Answers/diagrams were highly organized.

Senior Engineering Watchkeeper

QUALIFICATION BOARD ASSESSMENT SHEET

Candidate:	Submarine:	Date:

NOTE: Intent is for each assessor to ask one question per section.

Question		Content (0-8)	Conduct (0-2)	Score (S) (0-10)	Comments	
System Drawing	1					
Leadership/ Administration	2					
Stability and Liquid Handling	3					
General Technical Knowledge	4					
VCEOP with engineering scenario	5					
					Total =	
					RESULT (circle one) =	PASS / FAIL

Board Member Comments:

Board Member ID (Mandatory):

Name: _____ Signature:_____

BOARD Average Score	
AVERAGE: [(1)___ + (2)___ + (3)___ + (4)___ +(5)] / 5 =	

MAR TECH Senior Engineering Watchkeeper
Qualification Board Checklist

Candidate: _____ Board date: _____

Oral Board Prep: **a.** OJPR complete and reviewed
b. Board questions prepared
c. Score sheets prepared
d. Senior Watchkeeping Qualification prepared

Post Board Action: **a.** Score sheets signed, corrected and averaged
b. Senior Watchkeeping Qualification signed if member passed
c. Successful completion message drafted and released

COMPOSITION OF MAR TECH SENIOR ENGINEERING WATCHKEEPER BOARD	
Each board exam is to include 10 questions, of which only 5 are used, one per type. Where multiple questions are provided, the board may decide which to use prior to the board start. The question categories are laid out in the chart below. A board composition shall be IAW this QSP. The first and second question will be asked by the Board Chair. The VCEOP question will be asked by the MSEC. The Leadership/Administration question will be asked by the Board Chair. The remaining questions will be asked by the other board members to be determined by the Board Chairperson beforehand.	
Question types	**Quantity**
Engineering problem – drawing	1
Leadership/Administration	2
Stability and Liquid Handling)	2
General Technical Knowledge	2
VCEOP with Engineering Scenario	3